

# READING A BOUNDARY

BY

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RESEARCH AND THEORY

WHAT A SURFACE HOLDS, THE ALPHABET THAT READS IT, AND WHERE THE  
READING STOPS

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## WHAT A BOUNDARY HOLDS

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A closed surface holds a number of separable readings set by its own measure divided by the finest detail reaching it, raised to the dimension of the surface. Shape does not enter. Topology does not enter. Only the measure and the resolution do.

Two consequences follow. What a region holds scales with the area of its boundary and never with its volume. And a source at depth  $d$  inside a ball of radius  $R$  reaches harmonic degree  $l$  as  $(r/R)^l$ . Depth sets bandwidth, and a source prints a patch of angular size about  $d/R$  however small the source itself is. A spot on a boundary is always wider than the thing that made it.

**Note, 26 September.** The paragraph names the depth  $d$  and then writes  $(r/R)^l$  without defining  $r$ . Here  $r$  is the source's radius,  $r = R - d$ , the radius the next section uses.

### 1.1 DEPTH SETS BANDWIDTH

Under diffusion the boundary response to one interior source at radius  $r$  is the Poisson kernel, and its Legendre expansion carries  $(2l + 1)(r/R)^l$  at degree  $l$ . Everything else follows from that line. A source at the middle reaches degree zero alone and spreads over the whole sphere. A source against the shell reaches every degree and prints a small bright spot.

Reading the depth back out of a spectrum is a slope: dividing out the shape term and the conduction term leaves a straight line in degree whose gradient is twice the logarithm of  $r/R$ . Measured against sources placed at known depths, the recovery returns 0.3000, 0.5500, 0.8000 and 0.9200 for radii of 0.30, 0.55, 0.80 and 0.92. It holds under conduction as well, and that check caught an exponent being applied twice.

### 1.2 THE AREA LAW, COUNTED EXACTLY

Degrees up to  $l$  carry  $(l + 1)^2$  modes on an ordinary sphere, and the general count in  $n$  dimensions is a sum of two binomial coefficients. That count is a whole number with a formula. It needs no sampling and no rounding.

Weyl's law [50] predicts the same count from the surface measure alone: the volume of the unit ball in the boundary's own dimension, times the boundary's measure, times the eigenvalue to half that dimension, over two pi to that dimension. The two sides share no term. Compared at 2, 3, 5, 8, 12, 19, 24 and 40 dimensions, they agree to within 0.0122 percent, and at forty dimensions the exact count is a 157 digit integer.

The residual is a remainder, never a disagreement. At nineteen dimensions it runs 0.1131, 0.0282, 0.0070 and 0.0018 percent as the degree quadruples each time. An error term of order one over the degree falls by four for every quadrupling.

### 1.3 THE FLAT TORUS, AND WHY IT MATTERS

Spherical harmonics settle the dimension half of the claim, alone, because they live on spheres. A cube's surface has no such formula, which left shape resting on packing.

A flat torus has one. Glue opposite edges of a square and the eigenvalues of the Laplacian become  $(2\pi/L)^2$  times the squared length of an integer vector. Counting modes below a cutoff is counting integer points inside a ball. The count is an integer, it holds no transcendental, and it is available at any dimension by raising the every-square series to a power and adding up its coefficients.

This is worth having because a flat torus is not a sphere in any respect that could smuggle the answer in. It is flat where the sphere is curved. It has a hole where the sphere has none. Its symmetry group is a lattice where the sphere's is a rotation group. The same constant comes off both, exactly, with nothing sampled.

It also puts the remainder into known company. The difference between the exact lattice count and the volume of the ball is the lattice point problem, the Gauss circle problem in two dimensions and open in general. The leading term of the area law is settled here and the remainder is somebody else's open question.

**Note, 26 September.** "Open in general" overstates what is open. For balls in dimension  $d \geq 5$  the remainder is settled at order  $R^{d-2}$  (Walfisz 1957 [48]), and dimension four is settled up to a logarithm. Only dimensions two and three remain open. Cited from knowledge.

### 1.4 WHY SHAPE CANNOT ENTER

A seam where two edges meet contributes nothing to a surface measure. Neither do a cube's edges, its corners, or the point where three facets meet. Every one of them is measure zero on the surface. None of them can move a count that depends on measure alone.

Shape does not enter for that reason, and it is not a coincidence to be confirmed one shape at a time. A cube differs from a sphere precisely in its edges and corners, and those are exactly the parts that carry no measure. The count cannot see them. Overlap would be the exception, since overlap has positive measure and would count the same area twice.

The same rule governs the packing that measures this. Packed points may touch and may not overlap, and tangency is the densest limit. Touch freely, never overlap, and the measure is untouched either way.

### 1.5 DIFFUSION IS DWELL

Diffusion is not a blur applied to a picture. It is how long the light stayed inside before it left, and both readings come off one medium sorted by arrival.

Light that leaves early went nearly straight, still knows what it passed, and prints an edge. Light that leaves late has turned over so many times that it arrives from every direction at once. It fills the shadow it would otherwise have cast and carries how much in place of where. The same photons either way. A clock decides which of them are being counted.

That split is worth more than either half. A steady reading cannot separate absorption from scattering: a weakly absorbing dense medium and a strongly absorbing sparse one give the same number. Adding the clock breaks the degeneracy. The earliest arrivals give the shortest path and the straight-line geometry, the mean arrival gives the absorption, the spread of arrivals gives the scattering, and the tail gives the deepest excursions.

## 1.6 THE SMEAR

From inside, dwell is a tail behind a moving source. From outside there is no tail to see, because the inside is not what an outside observer has. What they have is the surface, and the tail arrives there as a smear.

The smear is worth more than the spot. A spot says where something is. A smear says where it was going and how long it took: its length is the dwell and its direction is the direction of travel, and both come off a surface that never moved.

## 1.7 THE CONDITIONS

The account needs four things and fails without any of them.

**Closed.** An open surface bounds no region. There is no interior to determine and flux leaves unaccounted. A surface pinched at isolated points and open between them fails on both counts: it is not a manifold at the pinch and it is still bounded everywhere else.

**Orientable, for containment only.** Weyl's law holds on any compact surface, orientable or not. The counting survives. What needs orientability is the outward normal, without which inside and outside are not defined. A non-orientable boundary keeps its capacity and loses its containment.

**A gradient.** No gradient, no flux. No flux, and the boundary records a uniform nothing however long it is watched.

**Coherence.** A hologram works by interfering a coherent field with a reference, which records phase as intensity on a two dimensional plate. Without coherence only intensity survives, and recovering the interior from intensity alone is the phase retrieval problem [6, 22]: non-unique in one dimension, generically unique and badly conditioned above it. Coherence is the difference between an inversion that is linear and a guess that is not.

Coherence also supplies the length the count was missing. The number of readings is unbounded as the resolution improves and finite at any resolution. The bound only becomes a number once something fixes a shortest length. In an optical setting the coherence length is that length, and the capacity is the area over the coherence area.

## 1.8 WHAT IS NOT ESTABLISHED

**Anisotropy.** The boundary map on an anisotropic medium is invariant under any diffeomorphism fixing the boundary [28], and a family of interiors then gives identical boundary data. Motion alone does not rescue this, since transforming the medium and its contents together leaves the boundary untouched. Motion under a law does: a diffeomorphism relabels space, straight lines curve, Keplerian orbits stop obeying Kepler, and only the untransformed world satisfies the law it was supposed to obey. Recovering orbital radii from a boundary light curve rests on exactly this, and the work is being done by Kepler and not by geometry.

**Stability.** Uniqueness is not stability. Recovering detail at scale  $x$  needs precision growing exponentially in  $1/x$ . Everything is recoverable in principle and buried under noise in practice. The spectrum of a real measurement dies by degree 46, and past there the inversion divides by a gain already below one part in a million.

**Shape above five dimensions.** Verified at three and five dimensions, where sphere, cube and cross-polytope give the same constant to within a factor of 1.14. Above that the count has to be packed, and packing stays honest only while the gap is well under the typical distance between two points on the surface. Holding to that limit at nineteen dimensions needs candidates in the hundreds of millions against a kept set in the hundreds of thousands.

**Note, 26 September.** The factor of 1.14 here and the spreads of 1.03 at three dimensions and 1.10 at five, in the section on the fourth defect, disagree. Nothing at orior d09b489 reconciles the two figures.

**The constant out front.** The scaling is provable and shape independent. The constant depends on packing density, which is known to be optimal in dimensions one, two, three, eight and twenty-four and open everywhere else. Dimension nineteen is open. The measurements here use a greedy packing, which reaches roughly half of the optimum and reaches it identically on every shape. It measures shape dependence and never packing efficiency.

**Note, 26 September.** The paragraph says the optimum at nineteen dimensions is open and also that the greedy packing reaches roughly half of the optimum. Where the optimum is open that fraction has no denominator, and at nineteen it cannot be measured. The optimum in three, eight and twenty-four dimensions was proved by Hales (2005) [21], Viazovska (2017) [45] and Cohn, Kumar, Miller, Radchenko and Viazovska (2017) [7], cited from knowledge.

## 1.9 THREE DEFECTS, AND HOW EACH WAS CAUGHT

Every one of these produced a plausible number and none announced itself.

The half-integer arm of the gamma function was written with  $(2j + 1)/2$  where it wanted  $(2j - 1)/2$ . It starts one step along and returns twice the right answer at every odd argument. The surface measure of every odd-dimensional sphere was doubled. The constant then alternated high and low with the parity of the dimension. The cube never touches the gamma function and stayed flat, which pointed at it. A count computed from the area and then checked against the area would have agreed with itself and found nothing.

**Derived, 26 September.** The formula as written is wrong. At odd  $n = 2k+1$ ,  $\Gamma(n/2) = \Gamma(k + \frac{1}{2}) = \sqrt{\pi} \prod_{j=1}^k (2j - 1)/2$ . Writing  $(2j + 1)/2$  over the same range gives  $\sqrt{\pi} \prod_{j=1}^k (2j + 1)/2$ , larger by the

factor  $2k+1$ : three at  $n=3$ , five at  $n=5$ . That factor is odd at every  $k$  and never two. The measure of  $S^{n-1}$  is  $2\pi^{n/2}/\Gamma(n/2)$ , and a gamma value too large makes the measure too small by the same factor. A doubled gamma halves the measure. No code at orior d09b489 carries this formula.

A limit on how wide a packing gap may be was written down and then ignored while picking gaps that would saturate quickly. The constants came back spread by three, seven and fourteen times at twelve, sixteen and nineteen dimensions, and read exactly like shape dependence. They were one gap being too wide, on every shape at once.

A remainder was tested for the wrong behavior. The error in a lattice count oscillates while its envelope falls. Requiring each radius to beat the one before it fails on a correct count. It did. Measured over a band of radii instead of at a point, the envelope comes down by a factor of forty-eight.

#### 1.10 THE DEMON, PRICED

Laplace's demon knows the state of everything and computes the rest. Quantum indeterminacy refuted it qualitatively. What the measurements here do is price the classical failure. That is the more useful statement.

A demon reading a boundary gets the measure over the resolution, raised to the dimension of the boundary. That is finite at any resolution and unbounded across resolutions. Nothing stops it improving except the cost of improving. The cost is exponential: recovering detail at scale  $x$  needs precision growing like the exponential of  $1/x$ , because the inversion divides by a gain falling as  $(r/R)^l$ . The spectrum of a real measurement dies by degree 46, and past there the division is by a number already under one part in a million. The demon meets no wall of principle. It meets a bill.

The distinction that matters cuts against the reading this work invites. Verifying the area law establishes that the channel is that wide. It does not establish that the interior holds only that much. Counted the ordinary way, a volume carries  $(R/x)^n$  degrees of freedom while its boundary carries  $(R/x)^{n-1}$ , leaving the boundary short by a factor of  $R/x$ . The holographic principle [1, 42] says the interior really is only boundary-sized and nothing is lost. That is a physical conjecture and nothing here tests it. The pipe was measured. What has to fit through it was not.

The fork is sharp. If the holographic principle holds, a demon at the boundary has exactly enough and its only difficulty is precision. If it does not, the demon is short by  $R/x$  however precise it becomes, and a counting deficit is not a rounding one. No arithmetic buys it back.

A third pressure applies either way. The demon sits inside the region it computes [53]. Its own state is written on the same boundary and counted against the same budget. A demon storing a description of the interior has to store it somewhere the interior can already see.

#### 1.11 READING THE DIMENSION OFF A LINE

A periodic structure in  $n$  dimensions, cut at an irrational angle and projected into fewer, comes out quasiperiodic: never one period, but several with irrational ratios between them. Penrose tilings are a five dimensional lattice seen in two. The correspondence runs both ways. Any quasiperiodic pattern lifts to a periodic lattice in high enough dimension.

**Note, 26 September.** The lift holds when the pattern has finitely many rationally independent frequencies. It is not a property of aperiodic order in general. The Penrose case is de Bruijn's (1981) [9], cited from knowledge.

That turns into a measurement. The count of rationally independent periods in a one dimensional reading equals the dimension of the lattice it was cut from. Cutting a square lattice at the golden slope and reading it back gives two gap lengths and never three, their ratio the golden mean to one part in ten to the fourteenth, their counts to four parts in ten thousand, no period at any length up to half the sequence, and every strong peak named by a pair of whole numbers against two generators. Against one generator the worst peak misses by 0.382 of it. Against two, by six parts in ten thousand.

**Note, 26 September.** The count equals the lattice's dimension when the cut is totally irrational, with the components of its direction independent over the rationals. A cut with a rational relation among them lies in a rational subspace of lower dimension, and the count reports that dimension. The golden slope in the square lattice is totally irrational, and the measurement above meets the condition.

A boundary reading can therefore report the dimension of a structure it never had access to, by counting how many of its periods refuse to be multiples of one another.

#### 1.12 A FOURTH DEFECT: THE WRONG QUANTITY HELD SMALL

The packing gap has to be small against the shape's own geometry. It was instead held to a fraction of the typical distance between two points on the surface, and in high dimensions those are not the same requirement.

The separation between two points drawn on a sphere concentrates near root two whatever the dimension is. A gap held at half of that is half the radius of curvature, and a cap that wide is nothing like the flat disc the count assumes it is. A cube's facets are flat and suffer none of it, and the two shapes part company and the answer reads as shape dependence. That criterion passed a twelve dimensional run whose constants spread by 2.1 times: sphere 5.37, cube 11.41, cross-polytope 8.20.

**Derived, 26 September.** The separation does not sit near root two at every dimension. On the unit 2-sphere the mean distance between two uniform points is  $4/3$ . On  $S^{n-1}$  the squared distance  $2 - 2\langle x, y \rangle$  has mean 2 and standard deviation  $2/\sqrt{n}$ , and the distance concentrates near  $\sqrt{2}$  only as  $n$  grows. Half of  $\sqrt{2}$  is 0.707 of the radius, above half of it.

Measured against the shape instead, at a gap well under every radius in play, the same three shapes give 0.664, 0.644 and 0.650 at three dimensions, 0.660, 0.619 and 0.640 at four, and 0.689, 0.627 and 0.658 at five. Spread 1.03, 1.07 and 1.10.

The shape half of the claim is verified there, and it is also the end of what packing reaches. Five dimensions took ten million candidates for twenty thousand kept points, and the count needed to stay in the flat regime grows the way a packing number does. No card makes that cheap at nineteen dimensions. The limit is structural and never a compute budget, and the exact routes on a sphere and on a flat torus are what reach up there.

## 1.13 THE NESTING, BOTH DIRECTIONS

Every boundary is inside another one, and everything a boundary holds is a boundary of its own. The outside of a shell is the inside of the next shell out, and the contents of a shell are shells with contents. Running outward: the cloud is inside its shell, the shell is inside the room, the room is inside whatever holds the room, and in this case what holds it is a universe. Running inward: each light is a closed surface, its own constituents are confined by it, and their traces smear across it exactly as the cloud's traces smear across the shell.

Neither direction terminates on its own, and both terminate for the reader.

Downward it stops at resolution. A constituent's smear on its own small surface subtends less than the finest detail the boundary between it and the observer can carry. Below that it is not there to be seen, and drawing it anyway would be inventing detail no boundary holds. The recursion stopping where resolution stops is the area law applied to the picture instead of stated about it.

Upward it stops at whatever the observer is inside. There is no vantage point outside everything, since standing outside one boundary places the observer inside the next. Stepping back from a wall does not escape a boundary; it changes which boundary is being read.

One level was measured. A source at depth  $r$  reaches degree  $l$  as  $(r/R)^l$ , the count on that boundary follows the area law, and the same constant comes off different shapes. That the arrangement repeats at every scale is a structural claim about the shape of the account and never a result taken from it. It follows if the same physics holds at every scale, and that premise was assumed and never measured.



---

## READING A LIT SET ON A BOUNDARY

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**Purpose:** Describe the three readings a lit set on a sphere admits, the floor under each, and what a caller has to know before lifting the module into another tree. **Scope:** `examples/00_blob_viz_tools/boundary_read.py`, `sphere_field.py`, `null_harness.py`, `arm_draw.py`.

`boundary_read.py` takes a set of lit points on a sphere and returns three readings of where those points sit, how they push the boundary, and how they twist it. Nothing in the module knows what a lit point means. The same code serves a hash state, a solar system or a file of bytes without changing. The caller supplies the meaning.

Every number below came from `python examples/00_blob_viz_tools/boundary_read.py --check` and can be produced again by running it.

**Note, 26 September.** Not every number below comes from that check. The degree table and the orthonormality figures come from `reading_rank.py` (DEGREES at :55) and `docs/arm-records.md:265-272`, the floors from `null_harness.py`, and the redraw row from `arm_draw.py`, all at orior ddeccb3. The Scope leaves out `reading_rank.py`.

### 2.1 WHAT HAS TO COME WITH IT

Two files and the standard library. `boundary_read.py` reaches into `sphere_field.py` at a single call, `legendre_column`, and `sphere_field.py` imports `math` and `sys`. There is no third dependency, no build step and nothing to install.

Two further files are worth taking and neither is required to run anything. `null_harness.py` measures the floor under each reading and imports both of the others. Lift it if the readings are going to carry thresholds, since the alternative is choosing them. `arm_draw.py` holds the arm record and several drawings of one arm set, and the harness imports it for the redraw floor. Lift that one if the reader is going to take arms as data.

### 2.2 PLACEMENTS

A placement decides where index  $k$  sits on the sphere. The choice shapes what can be read. Both are offered and neither is a default.

| placement                             | where index $k$ goes                                                | what it is good for                                 |
|---------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------|
| <code>golden_place(count)</code>      | height $1 - 2(k + 0.5)/count$ ,<br>longitude $ky$ [43, Corollary 2] | even coverage with no seam<br>and no pole pile [17] |
| <code>ring_place(rings, width)</code> | $rings$ latitudes of equal width,<br>$width$ longitudes on each     | a shift along a ring is a pure<br>rotation          |

`golden_place` carries the property the screw below depends on: one step in the index is one fixed rigid move, the same move everywhere along the spiral. `ring_place` puts its latitudes at the interiors of the bands. No ring lands on a pole where every point of it would pile into one place. Pick the ring placement when the operation under test is a rotation, and the golden placement when the operation is a shift of the index.

**Derived, 26 September.** The index shift  $n$  turns by  $n\gamma$  about the axis and slides the height by  $-2n/N$ . That is rigid in height and longitude, the coordinates the check tests, and it keeps area by Archimedes' hat-box theorem [23, *On the Sphere and Cylinder* I, Propositions 33, 42 and 43]. It is not an isometry of the sphere, because the axial radius  $\sqrt{1 - y^2}$  changes under the slide.

`as_angles(points)` converts either one into the colatitude and longitude pairs the readings want.

### 2.3 READINGS

All three readings come off one table. `complex_coefficients(angles, live, top)` sums the boundary coefficients of the lit set once, and the two harmonic readings take that table apart in different directions.

| reading      | call                                    | what it answers                                            |
|--------------|-----------------------------------------|------------------------------------------------------------|
| deflection   | <code>deflection(table, top)</code>     | how much power the lit set carries at each degree          |
| torsion      | <code>torsion(table, top)</code>        | the phase of the same coefficients, by degree and order    |
| octant share | <code>octant_share(points, live)</code> | what fraction of the set sits in each of the eight octants |

Moving every longitude by  $\alpha$  multiplies the entry at order  $m$  by  $e^{-im\alpha}$ . The magnitude holds still and the phase moves by  $m\alpha$ . Deflection reads the magnitudes and torsion reads the phases, and a turn moves the phases while leaving the magnitudes alone.

### 2.4 CHOOSING BETWEEN DEFLECTION AND TORSION

Deflection is blind to a rotation by construction [26]. Power per degree is a magnitude, and a magnitude does not record how the object was turned. Torsion measures the turn, and its sign gives the handedness.

**Derived, 26 September.** Deflection is invariant under every rotation, because a rotation acts on each degree by a unitary block [20, Section 2, eq. (15); Section 3, eq. (52)]. Torsion recovers only rotations about the polar axis, and those are the only ones the check tests.

The check moves a lit set along its rings by seven different amounts and reads both:

| reading    | what it did under a rotation                                        |
|------------|---------------------------------------------------------------------|
| deflection | moved by $2.5 \times 10^{-14}$ of itself, which is nothing          |
| torsion    | recovered the angle to $1.8 \times 10^{-13}$ radians, sign included |

The split is exact in both directions. The choice is not a matter of taste. A caller whose operations are rotations reads the twist. A caller wanting a quantity that holds still while the object turns reads the push. A caller who does not yet know which of those they have should read both and keep whichever carries a signal.

`turn_between(before, after, order=1)` performs the recovery, returning radians in the direction the longitudes run. Order one is asked by default because it pins the angle down without a wrap of its own; a higher order divides the angle and returns it only up to its own fraction of a turn. It returns `None` when neither set has a usable entry at that order, a real outcome and not an error to swallow.

## 2.5 THE OCTANT READING

`octant_share(points, live)` splits space into eight regions by the sign of each coordinate. Every region has a trilateral right-angle corner at the origin, all eight corners meet at that one point, and on the boundary the same split cuts eight congruent spherical triangles of area  $\pi/2$  each, by Girard [15, images f58 to f62], totaling  $4\pi$ . The measured shares sum to 1.0000000000000000.

The eight sign frames are not all the same kind of move. Their determinants split four at +1 and four at -1. Half are rotations of the first octant and half carry a mirror. A caller reading handedness off this split should know it is there before relying on it.

`octant_delta(before, after)` reports how much of the reading moved between two shares, as a percentage of the whole [29, Section 4.1, Proposition 4.2]. It is halved, because a share leaving one octant arrives in another and the two ends of a single move would otherwise be counted twice.

Eight regions is a small alphabet, and how small is a counting fact and not a judgment. The reading is rank 8, seven free numbers once the weight is fixed, and against a source with 256 degrees of freedom it moves in 8 directions and is blind in 248. A caller should know that number before building on this call, and should know that distinctness under such a reading is close to free: seven reals separate any few dozen arbitrary states whether or not the seven carry anything.

**26 September.** The sentence on seven reals separating a few dozen states is not derived here, a reading.

## 2.6 THE SCREW

`screw_of(count, amount)` returns the rigid move that shifting every index by amount comes to on a golden placement: a turn, an axial slide, and the pitch, the slide per unit of turn.

On 256 points the pitch is  $-0.003255258$  whatever the amount is. Across the six amounts the check tries, the spread between the largest pitch and the smallest is  $4.3 \times 10^{-19}$ . That is the floating

point grain and carries no dependence on the amount. One axis and one pitch therefore serve every shift, and a shift comes to a slide of the whole pattern along one fixed helix.

**Derived, 26 September.** The pitch is  $-2/(N\gamma)$ , and  $N = 256$  gives  $-0.003255258$ . `screw_of` computes the pitch from that definition (`boundary_read.py:213` at orior ddec3), and the spread of  $4.3 \times 10^{-19}$  is the rounding of the division. It says nothing about the placement. The move is rigid in height and longitude and is not an isometry of the sphere (the note under the placements).

`wrap_arm(count, amount)` names the indices a shift runs off the end of. They return as a second rigid arm, a full axial extent away from where the screw alone would put them. A shift is the screw plus that arm, and where a lit point lands is fixed once the amount is known.

## 2.7 WHAT IT COSTS

`complex_coefficients` is the expensive call. It walks the lit set once per order and computes a Legendre column per point. The work grows as the lit count times top squared. Degree eight over a few hundred lit points is immediate. A caller wanting a much higher degree should form the table once and hand it to both readings, the shape the module is already written in.

`legendre_column` climbs the diagonal and then recurs in degree, keeping every intermediate at unit scale [25]. Computing the same values from factorials overflows above degree about 150 and loses digits well before that, and it fails quietly: the low degrees stay right while the fine structure goes wrong. The recurrence is there to prevent a picture that looks plausible and is not.

**Derived, 26 September.** The two routes overflow at different degrees. The double's largest value is  $1.797 \times 10^{308}$ . The normalization's  $(l+m)!$  passes it at  $l+m = 171$ , degree 86 at  $m = l$  and degree 171 at  $m = 0$ . The unnormalized diagonal  $(2l-1)!!$  passes it at  $l = 151$ , since  $\log_{10} 299!! \approx 306.6$ . The figure of about 150 matches the second route, and the docstring (`sphere_field.py:93` at orior ddec3) names factorials, the first.

## 2.8 CHOOSING THE DEGREE

The degree is a choice about how much a reading carries, and not a limit the arithmetic imposes. The basis measures orthonormal to  $2.2 \times 10^{-14}$  at degree fifteen under exact quadrature [10], and to  $1.8 \times 10^{-14}$  at degree eight. Precision is not what holds a reading at any degree a caller is likely to want. The overflow above is a real ceiling and it sits an order of magnitude beyond that.

What does bound the reading is counting. A reading to degree  $L$  carries exactly  $(L+1)^2$  real numbers about its source, and against a source with  $N$  degrees of freedom it is blind in  $N-(L+1)^2$  directions at any precision whatsoever.

**Derived, 26 September.** The  $(L+1)^2$  real numbers count the coefficient table of a real function, where  $a_{l,-m} = (-1)^m \overline{a_{lm}}$  leaves  $(L+1)^2$  independent reals of the complex table's  $2(L+1)^2$ . Deflection itself carries  $L+1$  numbers.

| degree | coefficients | rank against 256 | blind | least singular value   |
|--------|--------------|------------------|-------|------------------------|
| 8      | 81           | 81               | 175   | 4.365                  |
| 12     | 169          | 169              | 87    | 3.818                  |
| 15     | 256          | 256              | 0     | $3.474 \times 10^{-3}$ |
| 16     | 289          | 256              | 0     | $1.524 \times 10^{-1}$ |

Pick the degree against the last column and never against the second. The smallest complete basis is degree fifteen, where  $(L + 1)^2$  first reaches 256, and it is the worst-conditioned usable one: the least singular value collapses to  $3.5 \times 10^{-3}$  there and recovers to  $1.5 \times 10^{-1}$  one degree higher, putting the worth of a single degree of headroom at 44 times in conditioning [54]. A caller picking the cheapest complete basis pays for that saving in precision afterwards.

## 2.9 THE FLOORS THESE READINGS SIT ON

Every reading here has a move that cannot change it, and running that move gives the reading's own grain in the units it reports. `null_harness.py` ships one such move per reading and reports the residual. A threshold is then measured and never picked.

| reading             | the move that cannot change it                          | residual                |
|---------------------|---------------------------------------------------------|-------------------------|
| deflection          | a rotation of the whole set                             | $2.539 \times 10^{-14}$ |
| torsion             | a rotation by a whole turn                              | exactly 0               |
| octant share        | a lit point relabeled inside its own octant             | exactly 0               |
| octant delta        | a reading against itself                                | exactly 0               |
| screw pitch         | changing the shift amount                               | $4.337 \times 10^{-19}$ |
| spectrum power      | a rotation of the sources                               | $4.005 \times 10^{-16}$ |
| a reading over arms | the arms redrawn as different shapes at the same weight | $2.220 \times 10^{-16}$ |

The last row is a null over the reader instead of over the object. An arm is identified by its topology together with its weight, and a shape is one realization of that. Drawing the arms differently while holding the weight cannot move a letter. `arm_draw.py` draws one set four ways, including as a line with a dwell particle on it, and no placement point changed arm under any of them.

Three of the seven are exactly zero and not merely small, because the moves are exact on integers. The octant reading has no continuum of small changes at all: a share is a count over a weight and both are integers, and at fixed weight the smallest change it can register is  $1/\text{weight}$ . Two states of *differing* weight can still give shares arbitrarily close together, leaving a caller who compares across weights to make a modeling choice about which states count as the same. Say that plainly instead of calling it a tolerance.

The companion measurement is not a null. Torsion recovers a known angle to  $1.803 \times 10^{-13}$  radians, which asks whether the reading moves the right amount when something did happen. A null asks whether it holds still when nothing did. A reading wants both answers and they are different questions.

## 2.10 RUNNING THE CHECK

---

command

---

```
python examples/00_blob_viz_tools/boundary_read.py --check  
python examples/00_blob_viz_tools/null_harness.py --check  
python examples/00_blob_viz_tools/arm_draw.py --check
```

---

The first holds the placement to the screw it should be, holds deflection and torsion to their opposite behavior under a rotation, and holds the eight octant shares to a sum of one.

The second measures the floors above, and before it reports any of them it hands itself a move that is not a null while declaring that it is, and a genuine null of the same shape. It has to catch the first and stay quiet on the second, and it withholds every floor and fails if either misbehaves. A checker that has never caught its own defect is a checker nobody has tested.

The third reports the redraw row at each of its three levels and holds four drawings that are not redraws against itself first. The second calls it. Running it separately is for reading the detail and not for coverage.

Exit status is zero when all of them pass. Running any of the modules without `--check` prints its own documentation, since all of them are libraries and have no work of their own to do.

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READING TRANSFORMS

---

**Purpose:** Write out every transform a boundary reading passes through, from the interior state to the pixel, with its action on the coefficients, whether it is diagonal in degree, what it leaves invariant, and what it costs. Keep the pieces in one table, leaving a transform already priced unpriced again while a transform never tried stays visible. **Scope:** `sphere_field.py`, `boundary_read.py`, `reading_rank.py`, `grid_error.py`, `natural_constants.py`, `room_view_template.html`.

## 3.1 THE OBJECT EVERY TRANSFORM ACTS ON

A reading is a real function on a closed boundary. Expanded in real spherical harmonics to degree  $L$ , it is  $f(n) = \sum_{l=0}^L \sum_{m=-l}^{+l} a_{lm} Y_{lm}(n)$ .

The reading is the vector  $a$ , of length  $(L+1)^2$ . At the ceiling the clock runs to,  $L = 10$ , that is 121 real numbers. Everything below is an operator on that vector.

Two facts about the vector before any operator touches it.

**Its length bounds what a reading can say.** A reading to degree  $L$  carries  $(L+1)^2$  real numbers about its source, whatever the source is. 256 sources read at degree 8 arrive as 81 numbers, and 175 directions of the source space are absent from the reading. The eight-letter octant alphabet is rank 8, absent in 248, with 7 free numbers left once the weight is fixed.

**Degree 15 dips and then recovers.** At full depth the least live singular value is  $3.47 \times 10^{-3}$  at degree 15 against 1.75 at degree 14. The exactly-complete degree is about 500 times worse conditioned than the incomplete degree beneath it, recovery arrives by 17, and saturation follows. Completeness of count and health of conditioning are separate properties.

**Note, 26 September.** The three values are in `docs/arm-records.md` at orior ddecbb3: 1.75 at degree 14 (:269),  $3.474 \times 10^{-3}$  at degree 15 (:270) and 2.52 at degree 17 (:272). The ratio of about 500 is  $1.75/3.474 \times 10^{-3} = 504$ , a ratio of least singular values. A condition number also divides by the largest singular value, which the table does not give.

## 3.2 THE TRANSFORM TABLE

*Diag* asks whether the operator is diagonal in degree: whether it acts on each degree  $l$  by one number, without moving power between degrees.

| #   | transform                                   | action on $a_{lm}$                                  | diag      | invariant               | co |
|-----|---------------------------------------------|-----------------------------------------------------|-----------|-------------------------|----|
| T1  | rotation, general [20, Section 2, eq. (15)] | $a_{lm} \rightarrow \sum_{m'} D_{m'm}^l(R) a_{lm'}$ | block     | $P_l$                   | On |
| T2  | rotation about read axis                    | phase moves by $-m\alpha$                           | yes       | $P_l$ , each $ a_{lm} $ | On |
| T3  | depth, outward carry                        | $a_{lm} \rightarrow (r/R)^l a_{lm}$                 | yes       | direction per degree    | On |
| T4  | conduction, smoothing                       | $a_{lm} \rightarrow e^{-l(l+1)\tau} a_{lm}$         | yes       | direction per degree    | On |
| T5  | pixel prefilter                             | $a_{lm} \rightarrow e^{-l(l+1)\sigma^2/2} a_{lm}$   | yes       | direction per degree    | On |
| T6  | band truncation                             | keep $l \leq L$ , zero above                        | yes       | every kept coefficient  | fr |
| T7  | parity, reflection                          | $a_{lm} \rightarrow (-1)^l a_{lm}$                  | yes       | $P_l$                   | On |
| T8  | screw, index shift                          | rotation $n\gamma$ , slide $-2n/N$                  | yes       | $P_l$                   | On |
| T9  | synthesis                                   | $f(n) = \sum a_{lm} Y_{lm}(n)$                      | no        | none                    | On |
| T10 | gradient                                    | $\partial_\theta, \partial_\phi$ by recurrence      | $l \pm 1$ | none                    | On |
| T11 | translation                                 | mixes every degree                                  | <b>no</b> | nothing per degree      | On |
| T12 | octant indicator                            | project onto 8 indicators                           | <b>no</b> | total weight            | ra |
| T13 | linear interpolation                        | not an operator on $a$ at all                       | n/a       | vertex values           | On |

T5 and T10 are not built. T1, T2 live in `boundary_read.py`; T3, T4 in `sphere_field.kernel`; T8 in `screw_of`; T12 in `reading_rank.octant_matrix`; T13 is the current viewer.

Three rows carry the weight of the table.

**T5 is not built and is one addition away.** Its action is T4's action with a different time.

**T11 is the exception proving the rule.** Translation is the only physical move in the table mixing degrees. It is avoided in the tree by giving each source its own depth factor in T3 instead of moving an origin.

**T13 is in the table to be seen for what it is.** It is not a transform of the reading. It is a straight line drawn between two samples of the reading, and the grid measurement below says how far that line sits from the curve it replaces.

### 3.3 THE COMPOSITION LAW

Rows T3 to T8 are all diagonal. Diagonal operators on the same eigenspaces commute, and composing them multiplies one number per degree. The entire chain from an interior state to a screen pixel therefore collapses to a single vector of  $L + 1$  gains:

$$a_{lm}(\text{screen}) = g_l a_{lm}(\text{state}), \quad g_l = (r/R)^l e^{-l(l+1)\tau_{\text{total}}}, \quad \tau_{\text{total}} = \tau_{\text{conduction}} + \tau_{\text{pixel}}$$

The two smoothing times add. The heat kernel on the sphere is a semigroup: smoothing for  $\tau_1$  then  $\tau_2$  is smoothing once for the sum. Nothing about that is an approximation.

**The whole physical and imaging chain is 11 multiplications.** At  $L = 10$ ,  $g$  is eleven numbers, recomputed when the depth, the smoothing or the ceiling changes and never per pixel.

**Derived, 26 September.** The pixel width  $\sigma$  varies per pixel (below, under the footprint), and  $\tau_{\text{pixel}} = \sigma^2/2$  enters  $g$ . The gains are then per pixel. Group the synthesis by degree,  $f = \sum_l g_l S_l$  with  $S_l = \sum_m a_{lm} Y_{lm}$ , and set  $q = e^{-2\tau}$ . Since  $g_l/g_{l-1} = e^{-2l\tau} = q^l$ , each degree past the first takes two multiplications, one for  $q^l$  and one for  $g_l$ . The filter costs one exponential and  $3L - 2$

multiplications per pixel, 28 at  $L = 10$ . Applying  $g$  to the whole vector costs 121 multiplications at  $L = 10$ , where the eleven count the gains alone.

**A correct pixel filter is free.** It is one addend inside a  $\tau$  already being applied. There is no second pass, no extra sampling, and no new operator.

### 3.4 WHY THEY ARE DIAGONAL, AND WHY ONE IS NOT

The table is not a coincidence and does not have to be memorized. The Laplace-Beltrami operator on the sphere has the degree  $l$  subspace as its eigenspace with eigenvalue  $-l(l+1)$ , that is,  $\Delta Y_{lm} = -l(l+1)Y_{lm}$ .

Any operator commuting with  $\Delta$  preserves its eigenspaces and therefore acts on each degree separately.

**T3, depth.** The field between the source and the boundary satisfies  $\Delta f = 0$  in the shell. Harmonic continuation outward is the solution of Laplace's equation there, and separating variables gives  $(r/R)^l$  as the radial factor. The exponent is the degree because the degree is the eigenvalue's index.

**T4 and T5, smoothing.** Both are  $e^{\tau\Delta}$ , the heat semigroup. An operator built as a function of  $\Delta$  commutes with it by construction.

**T1, T2, T7, T8, rotations and parity.** These are isometries of the sphere, and  $\Delta$  is built from the metric. They therefore commute with it. Each degree  $l$  subspace is an irreducible unitary representation of  $SO(3)$  of dimension  $2l+1$ , and rotation therefore mixes orders inside a degree while moving no power between degrees.

**Derived, 26 September.** T8 is not an isometry of the sphere. The index shift  $n$  turns by  $n\gamma$  about the axis and slides the height by  $-2n/N$ . That is rigid in height and longitude and keeps area, by Archimedes' hat-box theorem [23, *On the Sphere and Cylinder* I, Propositions 33, 42 and 43], but the axial radius  $\sqrt{1-y^2}$  changes under the slide. The slide moves power between degrees:  $Y_{10} \propto y$  goes to  $y - 2n/N$ , which has a degree 0 part. Only T8's rotation is diagonal. T8's entry in the diag column is no,  $P_l$  is not invariant under T8, and the diagonal rows at the composition law are T3 to T7.

**Derived, 26 September.** T7's  $(-1)^l$  is point inversion,  $n \rightarrow -n$ . A mirror through the equatorial plane gives  $(-1)^{l+m}$ .

**T6, truncation.** A spectral projection onto a set of eigenspaces.

**T11 does not commute with  $\Delta$ , and cannot.** A translation is not a map of the sphere to itself. There is no  $\Delta$  on the sphere for it to commute with. Re-expanding a field about a shifted origin mixes every degree into every other, at  $O(L^3)$  and with no per-degree number to carry it. Every cheap operation in this tree is diagonal, and the operation the tree avoids fails to be.

**Derived, 26 September.** Moving a source changes its moments triangularly. Degree  $l$  of the shifted expansion draws on degrees 0 to  $l$  of the old one, and degree 0 stays fixed: Greengard and Rokhlin's Lemma 2.3 [19, Section 2] gives the plane's case. T11 is not diagonal, and it does not mix every degree into every other. The  $O(L^3)$  cost holds by rotating, shifting along the axis and rotating back (Gumerov and Duraiswami [20, Section 1], crediting White and Head-Gordon [51]).

### 3.5 READING A PICTURE OFF THE COEFFICIENTS

#### 3.5.1 What a pixel asks for

A pixel does not ask for the field at a point. It asks for the field averaged over the solid angle it covers [38], and every antialiasing scheme is an attempt to answer that average.

**Supersampling estimates the average.** Take  $k$  samples inside the pixel and average them: cost scales with  $k$ , and the error falls about as  $1/k$ . It converges toward the answer and arrives at it only in the limit.

**A band-limited field has the average in closed form.** Convolution on the sphere is multiplication in degree. Averaging the field against a kernel  $K$  is therefore  $a_{lm} \rightarrow K_l a_{lm}$ , with no sampling anywhere. Choosing  $K$  to be the heat kernel of width  $\sigma$  gives T5, and one evaluation of the prefiltered field is the pixel's area average exactly.

**Derived, 26 September.** The heat kernel gives a Gaussian-weighted average. The exact average over a cap of angular radius  $\theta_0$  is also diagonal [14, Part 1], by Hecke's eq. (8) [24, Section 3]: with  $x = \cos \theta_0$ ,  $K_l = (P_{l-1}(x) - P_{l+1}(x))/((2l+1)(1-x))$ , and  $K_0 = 1$  with  $P_{-1} = 1$ . A square footprint is not zonal and is not diagonal.

The comparison is therefore not between one scheme and a cheaper scheme. Supersampling approximates a quantity this field hands over in one multiplication.

#### 3.5.2 The footprint and the time

The width to filter at is the pixel's own angular size on the boundary, available per fragment from the screen-space derivatives of the direction vector:  $\sigma$  is the length of the change in direction across one pixel, in radians, and  $\tau_{\text{pixel}} = \sigma^2/2$ .

Two properties follow that a mip chain [52] has to be built to imitate. The width varies per pixel, and the filter is therefore continuous in distance and in surface angle with no levels and no transitions between them. And the width enters  $\tau_{\text{total}}$ , where it is applied by the same eleven multiplications the chain already pays for.

#### 3.5.3 Where the degrees stop mattering

The prefilter also says which degrees can be skipped.  $e^{-l(l+1)\tau}$  falls below a threshold  $\epsilon$  once  $l(l+1) > \ln(1/\epsilon)/\tau$ , which gives  $l_{\text{max}} \approx \sqrt{2 \ln(1/\epsilon)/\sigma}$ . At  $\epsilon = 1/256$ , one level of an eight-bit channel,  $l_{\text{max}}$  is about  $3.33/\sigma$ .

**Stated honestly, this saving is small.**  $l_{\text{max}}$  reaches down to 10 only when  $\sigma$  is about 0.33 radians, which happens when the object spans a few tens of pixels. Above that size every one of the 121 terms contributes something visible and none may be dropped. The level-of-detail behavior is real, it is derived instead of authored, and it earns nothing at the sizes the viewer is normally used at.

**Derived, 26 September.** At the center of a disk of radius  $\rho$  pixels the direction changes by  $1/\rho$  radians per pixel.  $\sigma = 0.33$  gives  $\rho \approx 3$ , an object about 6 pixels across. The conclusion holds with more force.

3.5.4 *The grid against the coefficients, measured*

`grid_error.py` compares the drawn picture against the field the coefficients hold. The viewer samples 121 coefficients at 2,701 vertices of a  $36 \times 72$  grid, 5.0 degrees to a cell, and the rasterizer fills each cell with two planes.

Error per degree, for unit power in that degree alone, as a fraction of the picture's own contrast:

| degree | largest | typical | gradient median | gradient 95th |
|--------|---------|---------|-----------------|---------------|
| 1      | 0.0010  | 0.0004  | 1.25°           | 5.85°         |
| 4      | 0.0107  | 0.0026  | 3.22°           | 19.15°        |
| 7      | 0.0277  | 0.0058  | 5.34°           | 31.94°        |
| 10     | 0.0592  | 0.0108  | 7.47°           | 47.15°        |

The value error climbs as the square of the degree, as a linear interpolant must: interpolating a wave of length  $\lambda$  over a step  $h$  is wrong by about  $(\pi h/\lambda)^2/2$ , and a degree  $l$  harmonic has length  $360/l$  degrees.

**Derived, 26 September.** The largest error rises 59 times from degree 1 to degree 10,  $0.0592/0.0010$ , an exponent of 1.77. The square gives 2. With  $h = 5^\circ$  the estimate  $(\pi h/\lambda)^2/2$  gives 0.00095, 0.0152, 0.0466 and 0.0952 at degrees 1, 4, 7 and 10, within the table's rounding at degree 1 and above it at the other three. "Climbs as the square", a reading.

| field                       | largest | typical | gradient median |
|-----------------------------|---------|---------|-----------------|
| flat spectrum to degree 10  | 0.0312  | 0.0050  | 5.85°           |
| 63 deposits at $r/R = 0.80$ | 0.0273  | 0.0061  | 6.67°           |
| 63 deposits at $r/R = 0.60$ | 0.0174  | 0.0034  | 6.05°           |
| 63 deposits at $r/R = 0.40$ | 0.0075  | 0.0015  | 2.14°           |

The depth kernel is doing the work in that last column. A deeper reading carries less high-degree power by  $T_3$ , and the same grid draws it more accurately. The grid is wrong in proportion to the fine structure the state happens to hold.

**Read the gradient column.** The surface is shaded, and shading takes its normal from the gradient. A linear interpolant has a constant gradient inside a triangle. The drawn normal is a staircase across a field whose gradient turns smoothly. At degree 10 the median normal is 7.5 degrees off and one sample in twenty is more than 47 degrees off. That is the 5-degree quilt visible on the reconstruction, and it is a far larger error than the one percent in the value column.

A per-fragment evaluation has no interpolation error in either column, because there is no interpolation.  $T_{10}$  supplies the exact gradient in closed form for the same reason  $T_9$  supplies the exact value: the derivative of the expansion is another expansion.

3.5.5 *What the grid would cost to fix as a grid*

| grid             | vertices | largest | typical | gradient median |
|------------------|----------|---------|---------|-----------------|
| $36 \times 72$   | 2,701    | 0.0300  | 0.0051  | $5.83^\circ$    |
| $72 \times 144$  | 10,585   | 0.0077  | 0.0013  | $2.89^\circ$    |
| $144 \times 288$ | 41,905   | 0.0019  | 0.0003  | $1.46^\circ$    |
| $288 \times 576$ | 166,753  | 0.0005  | 0.0001  | $0.73^\circ$    |

Four times the vertices for a quarter of the error, the second-order convergence of a linear interpolant, confirmed at 3.95 against a predicted 4.

3.5.6 *The two working sets*

|                          | grid pass         | per fragment                |
|--------------------------|-------------------|-----------------------------|
| samples produced         | 2,701             | as many as there are pixels |
| basis                    | read from a table | built by recurrence         |
| bytes touched per pass   | 1.25 MB           | 484                         |
| samples at 900 px across | not applicable    | 636,172                     |

The grid pass is bound by a 1.25 MB table that does not fit in cache. A fragment evaluation reads the 121 coefficients and computes the basis from the direction. Its working set is 484 bytes at any sample count. **No frame time is claimed here.** These are sizes and counts; throughput belongs to a measurement that has not been made.

3.5.7 *Where this fails*

**Anisotropy.** The heat kernel is isotropic and a pixel's footprint near the silhouette is stretched. A single  $\sigma$  over-blurs across the short axis. The exact fix wants a directional kernel, which is not diagonal in degree and therefore leaves the composition law behind. The honest position is the same trade a trilinear filter makes against an anisotropic one.

**Edges that are not band-limited.** T5 is exact for a band-limited field, and the boundary field is band-limited by construction, since a reading is a truncated expansion. The room's walls, the beam stops and the arms have real corners, and prefiltering those rings. The split is therefore: the field gets the exact prefilter, the geometry keeps conventional coverage antialiasing. Applying T5 to the geometry would be a Gibbs artifact sold as a feature.

## 3.6 INVARIANTS, AND THE SPLIT THEY GIVE

$P_l = \sum_m a_{lm}^2$  is invariant under every element of  $SO(3)$  [26], because the degree  $l$  subspace is a unitary irreducible representation. Under a rotation by  $\alpha$  about the read axis,  $\arg a_{lm}$  moves by exactly  $-m\alpha$ .

**Derived, 26 September.** The coefficients are real (the object every transform acts on), and a real  $a_{lm}$  has no phase. Under a rotation by  $\alpha$  about the axis the pair  $(a_{lm}, a_{l,-m})$  turns by  $m\alpha$  as a

vector in the plane. Each  $|a_{lm}|$  changes, and  $a_{l,m}^2 + a_{l,-m}^2$  is invariant. The phase moving by  $-m\alpha$  holds for complex coefficients, and the T2 row's invariant is the pair's sum of squares.

The reading splits exactly along those two properties. **Power per degree is intrinsic**, a property of the source carrying no information about where the reader stands. **Phase is contextual**, a property of the source and the reader together, moving by a known amount under a known rotation. The split needs no estimator, and deflection and torsion are the two halves.

### 3.7 THE CONSTANTS, AND WHICH KIND THEY ARE

Every constant in the table is a computed number. `natural_constants.py` produces them to any requested length by integer arithmetic, verified at 60 places:  $\pi$ ,  $\sqrt{2}$ ,  $\sqrt{5}$ , the golden angle 2.399963229728653 ..., and the harmonic unit 0.282094791773878 ....

The golden angle is  $\pi(3 - \sqrt{5})$ , the index step of T8. The harmonic unit is  $1/(2\sqrt{\pi})$ , the degree zero harmonic and the scale every other one is built on.

**The verification carries no trusted digit string.** SHA-256's 64 round constants are defined as the first 32 bits of the fractional parts of the cube roots of the first 64 primes, and its 8 starting words the same for the square roots of the first 8 primes. Recomputing them from the primes and comparing against FIPS 180-4 [13, Section 4.2.2] checks 2,048 independently published bits. All 2,048 agree. The tool also grades every copy of the table in this tree against the definition: fifteen copies found, one of them the tool's own reference, and all fifteen match.

**Derived, 26 September.** The check covers the 64 round constants and the 8 starting words,  $72 \times 32 = 2,304$  bits (examples/proving/natural\_constants.py:338-350 at orior ddecbb3). The print at :273 says 2,048, the round constants'  $64 \times 32$  alone.

Pi is checked twice more, against a 50-digit prefix and against a second computation from Euler's arctangent identity. Machin's and Euler's identities share no term. Agreement between them is not two copies of one mistake.

**The measured value this produced.** The double the golden placement runs on sits  $4.441 \times 10^{-16}$  radians from the true golden angle. Over 63 indices the accumulated difference is about  $2.8 \times 10^{-14}$  radians, negligible, and now a number instead of an assumption.

#### 3.7.1 The other kind of constant

The 2022 CODATA adjustment [37, Table XXXII] lists each fundamental physical constant with a value and an uncertainty. About half read *exact*, because the 2019 redefinition of the SI fixed the speed of light, the Planck constant, the elementary charge, the Boltzmann constant and the Avogadro constant by definition. The rest carry an uncertainty no amount of computing will shrink: the fine-structure constant at  $1.5 \times 10^{-10}$  relative, the Newtonian constant of gravitation at  $2.2 \times 10^{-5}$ , every particle mass at its own figure.

**Note, 26 September.** The 2022 adjustment's Table XXXII gives the fine-structure constant at  $1.6 \times 10^{-10}$  relative. The Newtonian constant of gravitation at  $2.2 \times 10^{-5}$  holds.

**No constant of that kind appears in the reading path**, confirmed by search across every source file in the tree. The consequence is worth stating plainly: the reading engine carries no experimental uncertainty at all. Its only floors are chosen ones, and there are three.

| floor           | what sets it                           | how to move it               |
|-----------------|----------------------------------------|------------------------------|
| degree ceiling  | the reading's rank, $(L + 1)^2$        | raise $L$ , and pay $O(L^2)$ |
| number format   | double at 16 digits, shader float at 7 | carry more digits            |
| calibrated null | a move that changes nothing, measured  | a better null                |

The null is the only measured floor of the three, and it measures this tree's own arithmetic and not the world. The power-under-rotation null floor stands at  $4.005 \times 10^{-16}$ .

### 3.8 NOT SETTLED

**T5 and T10 are unbuilt.** The mathematics above is written and the shader is not. Nothing here is a frame time, and the claim that a fragment evaluation is faster than a table read is an architectural argument from working-set sizes, and not a measurement.

**The anisotropic case has no answer** beyond accepting the isotropic kernel's over-blur.

**The  $l_{\max}$  constant** of  $3.33/\sigma$  follows from a threshold of one level in eight bits. The crossover where dropping degrees begins to pay has not been measured against a real frame.

**T11's cost** is quoted as  $O(L^3)$  from the shape of the re-expansion and has not been timed here, because the tree avoids it.

**Depth and degree do not interact in the choice.** Depth multiplies a column by about a constant without reordering degrees. The price and the structure are chosen separately. Measured, and the mechanism is T3's diagonality.

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## THE OCTANT LEXICON

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**Purpose:** Record what an eight-letter reading of a lit set does and, at greater length, what it cannot do, including two conclusions withdrawn when the object being read turned out to be the wrong one. **Scope:** `examples/00_blob_viz_tools/octant_lex.py`, `quotient_coherence.py`, and SHA-256 compressing one block as the instance.

Eight regions, one letter each, and a word that a computation defines as it runs. The alphabet is general. It reads a lit set on a boundary and knows nothing about what lit it. The same eight letters serve any object that can be placed on a sphere.

Stated first, because a reader who takes one thing away should take this: the alphabet carries eight numbers about a source with 256 degrees of freedom, and the result that looks like a success is close to free. Neither of those is a defect in the tool. They are the size of the instrument, counted.

### 4.1 THE ALPHABET

Choose some regions. A state writes one letter per region, the letter being the share of the lit set sitting inside it. A step of the computation rewrites every letter at once, and the numbers tracked across the steps are the word it defines.

**The regions are free.** They may sit at any origin, inside the object or outside it, they may overlap each other and anything else, and they may take any shape. The same topology rules govern them as govern the rest of the dimensions, and shape does not enter: a region has to exist, and past that its geometry is a drawing convenience carrying nothing.

**Derived, 26 September.** On a placement a region reads through its membership vector, the set of points it holds. Eight regions holding the same points read rank 1, every letter equal. Geometry enters the reading exactly through membership, and a region's shape counts as far as it decides which points fall inside.

Two properties do follow from choices, and they are worth keeping apart from the reading itself. Regions tiling without gap or overlap force the letters to add to one, which costs one degree of freedom. Regions that overlap carry no such constraint and can therefore hold one number each.

### 4.2 THE INSTANCE THIS CHAPTER USES

Eight regions cut by the sign of each coordinate. Every one has a trilateral right-angle corner at the origin, all eight corners meet at that single point, and they tile space with no gap and no

overlap. On the boundary the same split cuts eight congruent spherical triangles, each with three right angles and each of area  $\pi/2$  [15], and the measured shares sum to 1.0000000000000000.

This is the convenient case and not the general one. It was picked because it is easy to state, easy to draw and cheap to compute, and every number below inherits the eight and the tiling from that choice.

Nothing above mentions a hash. What lit the set is the caller's business.

#### 4.3 THE SIZE OF THE INSTRUMENT

A reading to degree  $L$  carries exactly  $(L+1)^2$  real numbers about its source, whatever the source is. Set that against the number of degrees of freedom in the source and the shortfall is the blindness, before any measurement is taken.

| degree | coefficients | rank | blind | least singular value   |
|--------|--------------|------|-------|------------------------|
| 4      | 25           | 25   | 231   | 4.495                  |
| 8      | 81           | 81   | 175   | 4.365                  |
| 12     | 169          | 169  | 87    | 3.818                  |
| 15     | 256          | 256  | 0     | $3.474 \times 10^{-3}$ |
| 16     | 289          | 256  | 0     | $1.524 \times 10^{-1}$ |

Against a 256-bit lit set, a reading to degree eight recovers 81 directions and is blind in 175. The map reaches the rank its coefficient count allows at every degree below the source count. The shortfall in coefficients accounts for the blindness on its own. No sample size, no precision and no number of beams changes it.

The eight-letter alphabet here is rank 8, seven free numbers once the weight is fixed, blind in 248 of 256 directions.

Degree fifteen is the floor, where  $(L+1)^2$  first reaches 256. Reaching the floor is not reaching a usable reading: the least singular value collapses to  $3.5 \times 10^{-3}$  there and recovers to  $1.5 \times 10^{-1}$  at sixteen. One degree of headroom is worth 44 times in conditioning. A reading degree picked by counting coefficients alone lands on fifteen and reports success.

#### 4.4 THE OBJECT BEING READ

Everything from here down is one instance: SHA-256 compressing a single block, with its 256 working state bits taken as the lit set after each round. The instrument was tried on the hash, and the trying is not the subject.

A frame is the eight working words as the round leaves them, with nothing added back. That distinction decides every number below, and an earlier version of this page had it wrong.

The check counts which object it has instead of trusting the caller. A round only copies six of its eight words,  $b, c, d$  taking  $a, b, c$  and  $f, g, h$  taking  $e, f, g$ . On the working state all 378 carried words match the words they come from and on the digest none do.

## 4.5 THE PLACEMENT UNDER IT

The 256 state bits go onto a golden placement [46, 44]. One index of shift on that placement is a rigid screw: turn 2.399963 radians, axial slide  $-0.007812$ , pitch  $-0.003255258$ . The pitch holds for every amount. One helix carries them all, and where a shifted bit lands is fixed once the amount is known.

**Derived, 26 September.** The shift is rigid in height and longitude, the coordinates the pitch is read in, and it keeps area by Archimedes' hat-box theorem. It is not an isometry of the sphere, because the axial radius  $\sqrt{1 - y^2}$  changes under the slide. The boundary reading chapter carries the same note.

## 4.6 THE WORD

Compressing the empty message, read to degree eight. The delta [30] is the share of the reading that moved since the round before.

| round | octo  | oct1  | oct2  | oct3  | oct4  | oct5  | oct6  | oct7  | delta |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1     | 0.142 | 0.126 | 0.126 | 0.102 | 0.102 | 0.150 | 0.157 | 0.094 | 0.0%  |
| 2     | 0.122 | 0.174 | 0.122 | 0.078 | 0.130 | 0.113 | 0.113 | 0.148 | 12.9% |
| 3     | 0.171 | 0.111 | 0.120 | 0.137 | 0.103 | 0.103 | 0.145 | 0.111 | 14.0% |
| 4     | 0.111 | 0.103 | 0.128 | 0.120 | 0.154 | 0.120 | 0.103 | 0.162 | 12.8% |
| 8     | 0.156 | 0.117 | 0.141 | 0.133 | 0.117 | 0.078 | 0.102 | 0.156 | 8.5%  |
| 16    | 0.135 | 0.143 | 0.151 | 0.151 | 0.095 | 0.079 | 0.111 | 0.135 | 10.6% |
| 24    | 0.137 | 0.137 | 0.085 | 0.111 | 0.120 | 0.128 | 0.137 | 0.145 | 7.5%  |
| 32    | 0.095 | 0.127 | 0.143 | 0.095 | 0.127 | 0.143 | 0.151 | 0.119 | 10.1% |
| 48    | 0.136 | 0.127 | 0.127 | 0.085 | 0.127 | 0.102 | 0.153 | 0.144 | 12.4% |
| 64    | 0.150 | 0.129 | 0.114 | 0.121 | 0.129 | 0.114 | 0.143 | 0.100 | 5.9%  |

The digest at the end is e3b0c442 . . . 7852b855, the published value for the empty message [13]. That check confirms the trace is the hash and not something shaped like one.

## 4.7 THE DELTA AGAINST ITS FLOOR

The delta averages 11.9% over the first eight rounds and 7.1% over the last eight.

Two unrelated states of the same weight differ by 9.4% in this same alphabet. That figure is the redraw level, and `redraw_level` computes it by drawing 200 pairs of independent states at the matched weight and averaging the delta between them. Without it the delta is a number with no scale.

| quantity                    | value                          |
|-----------------------------|--------------------------------|
| redraw level                | 9.4%                           |
| delta, first eight rounds   | 11.9%, or 1.27 times the floor |
| delta, last eight rounds    | 7.1%, or 0.76 times the floor  |
| first round under the floor | 5                              |
| weight range across the run | 108 to 140 lit bits            |

The early rounds move the reading further than two unrelated states do. A reading whose delta clears its own independence level is not reporting counting noise, because counting noise is the level it cleared.

Two cautions on that, both of which bound it.

The fall is a trend in the means and not a monotone series. The per-round deltas over the first eight rounds run 12.9, 14.0, 12.8, 8.4, 11.9, 14.9, 8.5 and 12.0. Round 5 is the first crossing and not the last time the delta sits high.

Naming the move is not part of the measurement. A move carrying mass from octant to octant instead of mixing it will raise the delta above independence, and a rigid relabeling of the indices is such a move. Reading the early excess as the register shift is a reading of why, and it has not been measured. The measurement says only that the excess is present on the working state and absent on the digest.

#### 4.8 HOW MUCH OF EACH CHANGE THE ALPHABET COULD SEE

The delta says the reading moved. It does not say how much of the change the reading was able to move for, and those are different questions. A reading is a linear map, two states give the same letters exactly when their difference lies in its kernel, and the part of a change lying there is invisible however large it is.

The visible fraction answers the second question: the share of a difference lying in the row space [16] of the reading, which for eight indicator arms holds the per-octant totals of the change and discards the rest.

|                        | visible fraction | against an unstructured draw |
|------------------------|------------------|------------------------------|
| first eight rounds     | 0.0397           | 1.48 times                   |
| last eight rounds      | 0.0138           | 0.51 times                   |
| mean over all 63 pairs | 0.0278           | 1.04 times                   |

Read the split and never the mean. The early changes sit in the directions the alphabet can see above what chance would put there, the late changes sit in its blind directions below what chance would put there, and the mean is those two canceling. Taken alone the mean reports that the reading catches an arbitrary few percent, and that reading of it is an artifact of averaging.

The early to late ratio is 2.88, against 1.68 for the delta over the same split, on 8 pairs at each end. The direction is clear and the sample is small, and the ratio should be quoted with the 8 in the same breath.

**The two routes share no denominator, and one of them was raised as an objection to the other.** The delta divides counts by a weight moving 4.3 bits a round, and that moving denominator was offered as the reason the delta's trend might be bookkeeping instead of arrangement. The visible fraction is a ratio of two norms of one difference vector in raw counts and carries no such term. It shows the trend larger. The denominator was damping the effect and not producing it.

**Derived, 26 September.** The visible fraction is  $\|Pd\|^2/\|d\|^2$  (`quotient_coherence.py:15` at `orior ddeccb3`), a ratio of squared norms, and the control's 7/255 below is that squared share.

The control arm licenses reading any of this. Differences with no structure in them must return the fraction the row space predicts, and for differences holding the weight that figure is 7 of 255 and not 8 of 256: such a difference sums to zero, the all-ones direction lies inside the row space since the eight indicators add to it, and the difference lives in the hyperplane orthogonal to all-ones. Measured over 200 draws the control returns 0.0267 against a predicted 0.0275. `quotient_coherence.py` refuses to report the measurement at all when the control misses.

#### 4.9 THE READING THAT WAS CORRECTED

An earlier version of this page reported the alphabet as saturated: the delta sitting at the redraw level from round one, no room left to show mixing take hold, and eight letters called too coarse for the question. That was measured on the wrong object.

Frames came from `digest_after`, which adds the initial values back into the working state before returning. Every frame was therefore a finalized digest, and the feed-forward is a mixing step applied after the round, hiding the round's own structure. The count gives it away: on the working state all 378 carried words survive and on the digest none do.

The saturation was therefore a property of the digest sequence and not of the alphabet. `rounds_of` reads the working state by default now and keeps the digest to compare against. The coarseness statement was withdrawn, and the counting statement in the size of the instrument replaces it, since that one holds whatever object is read.

#### 4.10 THE DISTINCTNESS RESULT IS CLOSE TO FREE

Sixty-four rounds wrote sixty-four distinct signatures, a coherence of 100%, with no collisions. This is not evidence that the letters carry meaning, and it should not be presented as a finding.

The rank says why in one line. Seven free numbers separate sixty-four arbitrary states nearly always, whether or not the seven carry anything about the source. Distinctness here reports the resolution of a float and not a property of SHA-256, and any count of distinct signatures quoted without the rank beside it is quoting the float.

**Derived, 26 September.** A signature is eight counts over one weight, and distinctness follows from the counts with no float involved. At half weight each octant's count of about 32 points is hypergeometric with variance about 7.0,  $32 \cdot \frac{1}{4} \cdot 224/255$ . Two independent counts agree with probability about 0.106, seven free letters agree with probability about  $1.5 \times 10^{-7}$ , and the 2,016 pairs among 64 signatures expect about  $3 \times 10^{-4}$  collisions [12]. The same count derives the boundary reading chapter's sentence on seven reals separating a few dozen states.

It is computed because the complement would have been informative. A collision is a null permutation, a move this reading cannot see. None occurred, which rules that out and establishes nothing beyond it.

What would be evidence: a signature that predicts the round it came from.

#### 4.11 THE CHURN IS NOT STATIONARY

An earlier version of this page reported the arrangement moving at a constant rate from the first round to the last, and an earlier version of the check asserted first that the delta grows as the mixing takes hold and then that it holds level. It does neither. It falls, and it crosses the floor on the way down. Both readings were taken on the digest, where the feed-forward had flattened the run before the alphabet saw it.

#### 4.12 THE TWIST

Torsion is the reading that does carry position, and it does not bin anything. The twist between round 1 and round 64 reads 4.715210 radians. A phase recovers an angle to  $1.8 \times 10^{-13}$  radians while a magnitude reads a rotation as nothing at all.

**26 September.** `turn_between` (`boundary_read.py:156` at `orior ddeccb3`) assumes the second state is a rotation of the first. Rounds 1 and 64 are not, and the value it returns depends on which coefficient the search reads first. The  $1.8 \times 10^{-13}$  figure is for a true rotation. The twist of 4.715210 radians is a reading at most.

That figure sits under the same rank bound as everything else here. A phase recovers the turn it is given; it does not add coefficients to the reading.

#### 4.13 WHAT A FINER ALPHABET WOULD NEED

Refining the cells raises the coefficient count, and the counting table above is the bar it has to clear. A subdivision leaving more cells than there are points to write into them has bought dead symbols, and one reaching the rank floor has bought a reading whose conditioning collapses.

#### 4.14 RUNNING IT

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command

---

```
python examples/00_blob_viz_tools/octant_lex.py --check
python examples/00_blob_viz_tools/octant_lex.py --message "abc"
python examples/00_blob_viz_tools/octant_lex.py --read digest
python examples/00_blob_viz_tools/octant_lex.py --top 12
```

---

`--check` holds the trace to the published digest, holds the eight letters to a sum of one, counts the carried words to confirm which object it is reading, and holds the delta to clearing the floor early and falling across the run. `--read digest` reproduces the withdrawn reading, and its own check asserts that no carried word survives there.

---

A FINER ALPHABET, AND WHAT IT WOULD BUY

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**Purpose:** Work out what subdividing the octant alphabet would buy before anyone spends the compute on it, and record a derivation that did not hold along with what replaced it. **Scope:** the  $8 \times 4^k$  subdivision candidate, the rank bound, and 63 consecutive state pairs measured on both the working state and the finalized digest.

The eight-letter octant reading carries eight numbers about a source with 256 degrees of freedom. The obvious repair is a finer alphabet, and the candidate on the table is to subdivide each octant's spherical triangle [2, 18], giving  $8 \times 4^k$  cells at depth  $k$ .

### 5.1 THE ANSWER

Subdividing does not repair the reading, and the argument settling it is counting, and not statistics. The placement holds 256 points, and past a certain depth there are more cells than there are points to write into them [12].

| $k$ | cells | placement slots per cell |
|-----|-------|--------------------------|
| 0   | 8     | 32                       |
| 1   | 32    | 8                        |
| 2   | 128   | 2                        |
| 3   | 512   | 0.5                      |

At  $k = 3$  there are twice as many cells as placement points. Most letters can never be written at all, and the alphabet carries dead symbols. At  $k = 2$  the cell count equals the typical lit count, each cell holds two placement slots, and the reading has stopped being a coarse-graining: it is close to a permuted list of which bits are lit. On a 256-point placement  $k = 2$  is the last depth meaning anything, and by then the object being read is no longer a distribution over regions.

**Derived, 26 September.** The slots per cell in the table are means. Midpoint subdivision gives unequal cells: at  $k = 1$  an octant's center triangle has area 0.551 and each corner triangle 0.340, a ratio of 1.62, and the four sum to the octant's  $\pi/2$ . At  $k = 3$ , 256 points in 512 cells leave at least half the cells empty. The count gives at least half, and the claim of most needs more than the count.

Raising the placement's point count moves the ceiling but not the problem, since the redraw level below is set by the ratio of cells to lit points and the subdivision changes that ratio.

## 5.2 THE SAME CEILING, COUNTED ON THE READING INSTEAD OF THE PLACEMENT

The table above counts cells against placement points. The harmonic reading gives the same ceiling from the other side, and the two agree.

A reading to degree  $L$  carries exactly  $(L+1)^2$  real numbers about its source. Against 256 degrees of freedom, degree eight is rank 81 and blind in 175 directions, degree fifteen is where  $(L+1)^2$  first reaches 256, and the least singular value collapses to  $3.5 \times 10^{-3}$  there before recovering to  $1.5 \times 10^{-1}$  at sixteen. Refinement raises the count of numbers a reading carries; it does not raise the count past the source, and it makes the inversion worse on the way through the floor.

A finer alphabet is therefore bounded twice, once by how many cells the placement can fill and once by how many numbers the degree can carry, and neither bound is statistical.

## 5.3 THE SCALING EXPECTATION

Separately from the ceiling, a finer alphabet is expected to raise the delta and the redraw level together, leaving their ratio where it was.

The floor half of that is solid. Two independent states of the same weight give a level growing as the square root of the cell count, ordinary redraw statistics needing no assumption about the function being read. The delta half is only an expectation: a round's movement acts like a partial redraw and is expected to scale the same way. An attempt to derive that equality is recorded below, along with the reason it does not stand.

**Derived, 26 September.** The square-root growth holds while each cell holds many lit points, where the level grows as  $\sqrt{c/n}$  for  $c$  cells and  $n$  lit points. As the points per cell fall toward one, the level saturates at the set difference share, about 0.5 (the units note below), and stops growing.

## 5.4 A DERIVATION THAT DID NOT HOLD

The argument was that the delta equals the floor times  $\sqrt{s}$ , where  $s$  is the fraction of the lit set a round sends to fresh positions, on the grounds that points which did not move cancel cell by cell and leave the same calculation over  $sn$  points. The cell count then cancels between the two expressions and the ratio depends on  $s$  alone.

**The hole.** A share is a count divided by the weight, and the weight changes every round. On the working state it ranges from 108 to 140 lit bits and moves by 4.3 bits per round on average, 3.5% of the weight. A change in the denominator moves all eight shares even for points that never moved. The cancellation the argument rests on does not happen.

**The test.** Predicting  $s$  from the ratio squared and counting the lit points that actually moved, over all 63 consecutive pairs, with the digest column kept beside it because the earlier version of this note was measured there:

| quantity                                 | working state              | finalized digest |
|------------------------------------------|----------------------------|------------------|
| correlation of predicted against counted | 0.105                      | 0.068            |
| worst predicted $s$                      | 3.633, an impossible value | 3.508            |
| observed ratio, per pair                 | 0.382 to 1.906             | 0.507 to 1.873   |

As a per-pair estimator it fails on either object, and it returns values above one where no fraction can go. Changing the object moved every figure and changed no conclusion. Running it on both objects earned that much.

**What holds.** On the mean, with both sides in the same units, the law's central claim stays close. The counted relocation in the law's own units averages 1.0172, whose square root is 1.0085, against an observed mean ratio of 1.0361. The two agree to 2.7%.

The units matter and were the source of an earlier disagreement about this: a set difference between two states of weight  $n_2$  out of  $N$  saturates at  $1 - n_2/N$ , about one half, and never at one, and a raw set difference of 0.5 already means complete independence. Comparing  $\sqrt{0.5} = 0.707$  against a ratio near 1.0 compares two different quantities.

The mean prediction therefore survives and the per-round estimator does not. Neither outcome supports using this argument as the reason a finer alphabet fails, and the two ceilings above are the reasons to give.

## 5.5 THE CONSTANT THAT REPLACED IT

The counted relocation is close to constant, the useful finding to come out of the test.

| quantity                                             | working state                     | finalized digest |
|------------------------------------------------------|-----------------------------------|------------------|
| set difference between consecutive states            | mean 0.5215, range 0.421 to 0.619 | mean 0.4969      |
| the same in matched units, where 1.0 is independence | 1.0172                            | 0.9976           |
| bits differing between consecutive states            | mean 129.9 of 256                 | mean 127.7       |
| what a coin would give                               | 128.0                             | 128.0            |

Every round relocates the lit set about as completely as an independent draw would, from the first round onward, with the working state sitting 1.7% above independence and the digest 0.2% below it. Reading  $s$  off the ratio is therefore estimating something that barely varies, and the backwards reading carries correspondingly little.

An earlier version of this note put worked examples in a table,  $s = 0.45$  at round 1 to 2 and  $s = 0.91$  across the first eight. Both were wrong, and they were wrong in a way worth naming: they came from reading the ratio off aggregate percentages instead of per pair, and they bracketed the true value by accident.

## 5.6 WHAT A FRAME ACTUALLY IS

Worth stating, because it bounds every result above, and because an earlier version of this section had it backwards.

A frame is the eight working words as the round leaves them, with nothing added back. `rounds_of` reads that by default. It can also return the finalized digest under `--read digest`, where `digest_after` adds the initial values back into the working state before returning, and that feed-forward is a mixing step applied after the round.

The two objects are told apart by counting and not by assertion. A round only copies six of its eight words [13],  $b, c, d$  taking  $a, b, c$  and  $f, g, h$  taking  $e, f, g$ . Over sixty-three transitions that is 378 carried words, and on the working state all 378 match the words they come from while on the digest none do.

That count also settles a question this section previously recorded as closed the wrong way. The register shift is present in the working state exactly, since being a register shift consists of matching all 378 carried words. The earlier text reported the shift as invisible, on a test that undid the word shift and found 108.9 of 224 bits differing against a coin's 112.0 with no exact match in sixty-four rounds. That test was run on digest frames, and it was measuring the feed-forward.

What is not established is a link between the shift and the octant delta. The delta's early excess over the redraw floor is measured, and a rigid relabeling of indices is the kind of move producing such an excess, but attributing the excess to the register shift is a reading of why and has not been measured. The two statements sit at different levels and only the first is a count.

## 5.7 WHAT WOULD SETTLE THE SUBDIVISION

One run at  $k = 1$  and  $k = 2$ , each with its own redraw level computed at the matched weight, reporting the ratio and not the delta. Two ratios near the  $k = 0$  value confirm the expectation. A ratio falling with  $k$  refutes it cheaply, and the two ceilings still stand either way.

**Ownership.** The subdivision candidate and the run are the engine's. The refutation of the  $\sqrt{s}$  derivation is the engine's, and the rank bound is the engine's. The reproduction, the units correction, the object correction and the reruns above belong to this document.

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THE SHIFT REGISTER, READ AS A DEFORMING SURFACE

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**Purpose:** Record what the deformation of a harmonic reading of the SHA-256 compression state is, prove it belongs to the state and not to the observer or the probe, and trace it to the one structural feature of the compression function that could produce it. **Scope:** `examples/proofing/state_deflection.py`, and the figures in `theory/figures/theory/boundary/state-deformation/`.

A harmonic reading of the SHA-256 compression state renders a surface, and that surface deforms as the round clock turns. The observer-invariance making the reading trustworthy is stated first, because it is the calibration letting the rest be read as measurement and not as viewpoint.

### 6.1 THE OBJECT

SHA-256 compresses in sixty four rounds over a state of eight thirty-two bit words, which is 256 bits. The reading places those 256 bits on 256 points of a ring arrangement and expands the lit set in spherical harmonics. The state becomes a field on a sphere, and the field has a shape. Bit  $i$  lights point  $i$ ; no mapping was invented to make the instrument fit the object, because 256 bits and 256 points are the same count.

**26 September.** The order taking bit  $i$  to point  $i$  is a choice. The placement is ring-major (`ring_place`, `examples/00_blob_viz_tools/boundary_read.py:71`) and the lit set is word-major (`lit_of`, `state_deflection.py:110`, both at orior ddeccb3), and together they put word  $s$  on ring  $s$ . The notes under the mechanism and the scope depend on that. That no mapping was invented is a reading.

The compression agrees with the published digest of the empty string before any of this is read. The object under the instrument is SHA-256 and not an approximation of it.

### 6.2 THE CALIBRATION: WHICH PART OF THE SHAPE IS THE OBSERVER'S

A shape read this way has two anisotropies, and they answer to a rotation of the observer differently. This is standard representation theory of the rotation group, stated here because it is the null the measurement rests on, and not because it is new.

**Deflection**, the power per degree  $P_l = \sum_m |a_{lm}|^2$ , is invariant under every rotation in  $SO(3)$ : the degree- $l$  subspace carries a unitary irreducible representation of the group [11], and a magnitude built from it cannot record how the object was turned [26]. Measured null under a whole-step rotation:  $2.539 \times 10^{-14}$ , the arithmetic and not the object.

**Torsion**, the phase  $\arg a_{lm}$  of the same coefficients, moves by exactly  $-m\alpha$  under a rotation by  $\alpha$ . It records the turn, sign included. Measured recovery of a known angle:  $1.803 \times 10^{-13}$  radians.

**Derived, 26 September.** The phase moves by exactly  $-m\alpha$  under a rotation about the polar axis. A rotation about any other axis mixes the orders  $m$  within each degree, and one coefficient's phase no longer records the turn alone.

The magnitude of the anisotropy is therefore observer-independent and its direction is observer-dependent, and the split is exact. This is a property of the reading and holds for any lit set, SHA-256 or otherwise. Its role here is precise: a change in the shape's **magnitude** across the clock is a change in the object, while a change in its **direction** could be either the object or the frame. Every claim below is made on a magnitude, where the observer has been divided out, or is checked against a control holding the frame fixed.

### 6.3 THE MEASUREMENT, ON THE SURFACE ITSELF

The reading renders exactly one deformed surface in the scene, and it is the state's. Every enclosing shell measures a vertex-radius standard deviation of exactly zero, and each is a perfect sphere. The state's surface measures a standard deviation of 0.999 against a mean radius of 8.795, at 11.4 percent. Read as a radius field over the sphere and tracked across the sixty four rounds:

| feature                 | degree | across the clock                                                     |
|-------------------------|--------|----------------------------------------------------------------------|
| the elongation, a lemon | 2      | $q_{zz} \approx -1.50$ at every round, 17 percent of the mean radius |
| the offset, a cone      | 1      | $d_z$ swings about 0.07 and changes sign several times               |

The quadrupole is a fixed feature: the surface is elongated by the same amount, in the same axis, the whole way through. The dipole migrates and inverts. The cone walks around the surface and changes which pole it favors more than once.

### 6.4 THE CLAIM, STATED AS A NUMBER

The dipole does not merely move. It moves **coherently**: its sign, taken round by round, comes in runs, and not in the alternation independent signs would give. Twenty runs against an expected 30.5, at a run-length test assuming nothing about the values themselves:

$$z = -2.87$$

Measured a second way, on a different object through different code, the rendered mesh in the viewer against the lit set in Python, the same statistic reads  $z = -3.0$ . Two pipelines with no shared arithmetic land two hundredths apart.

**Derived, 26 September.** A value printed to one decimal as  $-3.0$  lies between  $-3.05$  and  $-2.95$ , and it sits 0.08 to 0.18 from  $-2.87$ . The printed figures show agreement within two tenths. Two hundredths does not follow from them.

A surface computed from a slowly changing state drifts at that level. The next section is the control pinning the change to the state.

## 6.5 PINNING THE CHANGE TO THE STATE

Two interventions, because the round clock advances everything at once and an observation that the picture changes when the clock moves does not by itself say what moved it.

**The probe was moved and nothing happened.** Holding the round fixed and swinging the beam 180 degrees in azimuth, 128 in elevation and 4.7 times in radius left the surface unchanged in every digit: mean radius, dipole and quadrupole all identical before, during and after. One round of the clock, with the beam untouched, then moved the dipole by five times. The surface answers to the state and not to the probe.

**The rounds were replaced by independent states.** The real round sequence gives  $z = -2.87$ . A control swapping the rounds for independent states of the same per-round weights, everything else identical, gives  $z = 0.00$  with a spread of 0.94. The real sequence sits 3.1 control deviations out. The drift is in the succession of states, and not in the reading of any one of them.

## 6.6 THE MECHANISM, WHICH IS NOT EXOTIC

SHA-256's round computes two words and shifts the other six:

$$\begin{array}{ll} a' = T_1 + T_2 & e' = d + T_1 \\ b' = a & f' = e \\ c' = b & g' = f \\ d' = c & h' = g \end{array}$$

Six of the eight words at round  $r$  are round  $r - 1$ 's words in a new position, measured at 63 of 63 rounds. A surface computed from a state that is five-sixths carried over cannot jump between consecutive rounds, and a coherent, autocorrelated drift is therefore the shift register showing.

**Derived, 26 September.** Six of the eight words carried over is three quarters of the state. Five-sixths does not follow from the table.

**Derived, 26 September.** The run count follows from the shift with a number. The dipole is zonal (zonal\_dipole, state\_deflection.py:274 at orior ddec3), and word  $s$  sits on ring  $s$  at colatitude  $\pi(s + 1/2)/8$  (the note under the object). A fresh word enters at  $a$  or at  $e$  and rides four rings over four rounds. With  $A_r$  and  $E_r$  the popcounts of the two fresh words at round  $r$ , and  $c_j = \cos \theta_j = 0.981, 0.831, 0.556, 0.195$  for  $j = 0, \dots, 3$ , the dipole is  $d(r) = \sum_j c_j A_{r-j} - \sum_j c_{3-j} E_{r-j}$ . Taking the fresh popcounts independent with equal variance, its correlations at lags 1, 2 and 3 are 0.693, 0.354 and 0.096, and zero beyond. By Sheppard's rule [41] a sign change between neighbors has probability  $\arccos(\rho_1)/\pi = 0.256$ , about 17 runs in 64, against 20 measured and 30.5 for independent signs. The  $z = -2.87$  reproduces from 41 and 23 signs of 64: the expected run count is 30.47 and its standard deviation 3.65 (Wald and Wolfowitz [47]).

**That is the claim entire, and it is worth stating without deflation.** The reading was built to show a field. Nobody built it to expose the Merkle-Damgard shift register, and the register turned up in it anyway: legible to the eye as a lemon whose cone walks and inverts, and holding at three standard deviations when the drift is measured against a control. The result is not a new property of SHA-256. It is a demonstration that a harmonic boundary reading makes a known structural property of the compression function visible and measurable, with the observer divided out and the cause established by intervention.

**Note, 26 September.** Merkle–Damgård names the chaining of compression calls across message blocks [36, 8]. One block’s compression holds none of it. The word shift in the table above is SHA-256’s step function [13, Section 6.2.2].

## 6.7 SCOPE, STATED PLAINLY

**The observer-invariance is classical.** Rotation-invariant power and orientation-carrying phase are standard  $SO(3)$  representation theory, the same fact behind an angular power spectrum being usable without its phases. It serves here as the calibration and finds nothing new.

**What is specific to SHA-256** is the behavior of the anisotropy across the clock: a persistent quadrupole, a dipole inverting in runs at  $z = -2.87$ , and the trace to the shift register by intervention.

**Derived, 26 September.** The placement predicts the persistent quadrupole, for random states as much as for SHA-256’s. With word  $s$  on ring  $s$  (the note under the object), the zonal quadrupole is proportional to  $\sum_s (3 \cos^2 \theta_s - 1) \text{pop}_s$ . The ring coefficients are 1.886, 1.074,  $-0.074$  and  $-0.886$ , mirrored in the southern half, and they sum to 4.0, where the sphere’s area average of  $3 \cos^2 \theta - 1$  is zero. At a mean popcount of 16 the expected term is 64 units. With a popcount variance of 8 per word the fluctuation is about 9.4 units, and the sign holds at every round with a wobble near 15 percent. The `--shape` run scores the degree-2 power against shuffled lit sets on the same placement (`_shape`, `state_deflection.py:366` at `orior ddecbb3`). That control separates SHA-256 from the placement, and this chapter does not report its score.

**No claim is made** about a weakness in SHA-256, about its output, or about anything a cryptographic reduction would touch. The reading is taken on the intermediate compression state, which carries the shift structure by design. The digest, being the feed-forward of all sixty four rounds, is a different object and is not read here.

**Note, 26 September.** The digest is the state after round 64 with the block’s chaining input added word by word [13, Section 6.2.2, step 4]. The feed-forward is that one addition, made once after the last round.

## 6.8 REPRODUCING IT

| command                                  | what it returns                       |
|------------------------------------------|---------------------------------------|
| <code>state_deflection.py --check</code> | the instrument, against the digest    |
| <code>state_deflection.py --shape</code> | dipole and quadrupole per round       |
| <code>state_deflection.py --runs</code>  | the autocorrelation against a control |

The figures, and the intervention table, are in `theory/figures/theory/boundary/state-deformation/`.

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ROUND COUNT AS DEPTH

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**Purpose:** Specify an experiment measuring how fast the interior stops reaching the boundary as a function of how many rounds it passed through, and reporting where that curve crosses below the detection floor. **Scope:** `src/engine/c/sha256/bench/`, the representation arm's exact contract, and a driver that does not yet exist.

**This chapter specifies work and does not report it.** Everything below is a design: the observables are built, the driver is not, and no curve has been measured. It is transcribed with the rest because a specification written down before the run is the only kind that can be checked against the run afterwards.

**Note, 26 September.** The observables were built as the benches in `src/engine/c/sha256/bench`. Commit `bdaed61` deleted that directory on 24 September, 54 files, and none of them is in a file at `orior 1948ae1`. At the pin the observables are not built, and the driver is not built.

## 7.1 THE IDEA IN ONE LINE

The boundary work measured information reaching a surface as a function of depth, and found it falls as  $(r/R)^l$ . Here depth is the round number. The question is the same one: how much of the input reaches the output, measured against how far in it started.

## 7.2 WHY THIS IS THE RIGHT QUESTION AND BREAKING IT IS THE WRONG ONE

A break needs a boundary observable departing from flat by more than sample noise, at full rounds. The tree's own instruments say there is none:  $H(\text{digest} \mid \text{input}) = 0$ , which leaves the bias bound limited by sample size alone with no floor beneath it, and the keyhole, SAC [49] and Renyi [39] benches read flat at full rounds. Twenty years of linear [34] and differential [4] cryptanalysis reach about 31 of 64 rounds and no further. Hunting the full-round hole is hunting a thing the detector already says is not there.

**Derived, 26 September.** Every deterministic function has  $H(\text{output} \mid \text{input}) = 0$ , biased ones included. The equality holds for a biased function as much as for a flat one, and it bounds no bias.

**Note, 26 September.** Thirty-one steps is the collision record: Mendel, Nad and Schl  ffer (2013) [35], with a colliding pair found in practice by Li, Liu, Wang, Dong and Sun (2024) [32]. Other attacks reach further. Semi-free-start collisions reach 39 steps (Li, Liu and Wang, 2024 [31]), and the biclique attacks of Khovratovich, Rechberger and Savelieva (2012) [27] reach 45 steps for preimages and 52 for pseudo-preimages. The two biclique step counts are cited from knowledge.

Measuring the decay is different. Reduced rounds leak, everyone knows they leak, and nobody has written down the leak as a curve in the depth variable with a detection floor drawn on the same axes. That curve is a real result. It quantifies the difficulty instead of asserting it, and it makes a falsifiable claim: the leak at round  $k$  is this size, and here is where it falls under what this many samples can see.

### 7.3 WHAT IS MEASURED

Run SHA-256 truncated at  $k$  rounds, for  $k$  from 1 to 64, and read a boundary observable at each. The observable is one already built:

**Strict avalanche.** Flip one input bit, count how often each output bit flips, over many inputs. A flat function sits every counter at half the sample size. The departure from half, summed across output bits, is the leak. This is the keyhole scan with the round count as a parameter.

**Bit bias.** How far each output bit's one-rate departs from half. The survey bench, truncated.

**Collision entropy of the output byte histogram.** The Renyi bench, truncated.

Each returns a single number per round count. The run is one curve per observable, 64 points each.

### 7.4 THE DETECTION FLOOR, DRAWN ON THE SAME AXES

A bias of size  $b$  needs about  $1/b^2$  samples to see above the noise.  $N$  samples can see down to about  $1/\sqrt{N}$ . That floor is a horizontal line on the same plot as the decay curve, and where the curve crosses under it is the round count past which this run cannot tell the function from flat.

**Derived, 26 September.** At a flat function one counter's rate over  $N$  samples has standard deviation  $1/(2\sqrt{N})$ . The strict avalanche observable sums |departure| over  $M$  counters, and that sum has a null mean of  $M\sqrt{2/\pi}/(2\sqrt{N})$ ,  $M$  times the mean of a half-normal. A summed observable is plotted against that line. The  $1/\sqrt{N}$  above is the floor of one counter, up to a factor of two.

The floor is not a nuisance to be hidden. It carries the result: it says what was in reach of this much compute, and a later run with more samples can lower the line and show the curve continuing underneath, or show it genuinely flattening.

### 7.5 THE EXACT-ARITHMETIC ROLE

The decay spans many orders of magnitude, from a large leak at a few rounds to nothing measurable at full rounds. Read in single precision the small end is noise about noise. The representation arm's exact contract carries the small counts without rounding. The curve stays a curve down to where sample size and not arithmetic stops it. That separates measuring the decay from measuring the floating-point floor, and it is the reason this experiment belongs in this tree and nowhere else.

**26 September.** The counts are integers. A 64-bit counter holds them exactly to  $1.8 \times 10^{19}$ , and single precision holds every integer exactly up to  $2^{24}$ . Exact small counts need only an integer counter. That this experiment belongs in this tree and nowhere else is a reading.

## 7.6 WHAT EACH OUTCOME WOULD MEAN

**The curve falls smoothly and crosses the floor around the round count cryptanalysis already reaches.** The expected result. It turns the known reduced-round weakness into a measured decay rate and puts a number on the margin.

**The curve has a step, a shoulder, or a bump that does not fall.** A round range where more structure survives than the smooth decay predicts. That is where a cryptanalyst would look, and the framework would have pointed at it instead of guessing. Almost certainly it will not happen at full rounds, and the experiment would show it if it did.

**The curve is already under the floor by a few rounds.** The samples were too few. Lower the line with more samples and run again. Nothing is concluded from a curve starting under its own floor, and a run passes that check before its flat tail means anything.

## 7.7 WHAT THIS DOES NOT CLAIM

It does not claim a break, a bias at full rounds, or a shortcut to a preimage. It claims a measurement: the leak as a function of depth, with the floor saying how far down the measurement could see. A flat tail under a floor is not evidence the function is flat there. It is evidence the run could not see a leak that small, a statement about the run and not about SHA-256.

## 7.8 RUNNING IT

The benches are in `src/engine/c/sha256/bench`. This experiment is a driver calling them with a round-count parameter and collecting the curve, plus a plot through `build_chart_view.py` with the floor drawn as a second series. The one piece of new engine work is truncating the compression function at an arbitrary round, which the early-exit path in `sha256_core.c` already does for the anchor and would be extended to stop at any  $k$ .

**Note, 26 September.** The Scope and this section contradict the note under the design statement. At orior d09b489, as at 1948ae1, `src/engine/c/sha256/` holds no file and no `sha256_core.c` is in the tree. The benches, the early-exit path and the truncation this section names are not built.



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## BIBLIOGRAPHY

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Every measurement in this work is built on somebody's published result, and the first purpose of this chapter is that those people are credited by name. A surname in a footnote is not a citation. The second purpose is that the credit is checkable: each source is registered in `orior_citations`, where the copy sits itself, and this bibliography is generated from that registry and pinned to one commit of it. A reference therefore resolves not to a title but to exact bytes.

Credit here does not depend on agreement. Where this work reproduces a published result it says so, and where it fails to reproduce one it says that too, under the same names, because a refutation rests on the original as much as a confirmation does.

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A reader who disagrees with a number in this research paper can ask which copy it was checked against, and the hash answers exactly. Two copies of a paper are rarely the same bytes.

- [1] G. 't Hooft. *Dimensional reduction in quantum gravity*. arXiv:gr-qc/9310026. 1993.
- [2] J. R. Baumgardner, P. O. Frederickson. *Icosahedral discretization of the two-sphere*. SIAM Journal on Numerical Analysis 22, 1107-1115, doi:10.1137/0722066. 1985.
- [3] J. D. Bekenstein. *Universal upper bound on the entropy-to-energy ratio for bounded systems*. Physical Review D 23, 287-298, doi:10.1103/PhysRevD.23.287. 1981.
- [4] E. Biham, A. Shamir. *Differential cryptanalysis of DES-like cryptosystems*. Journal of Cryptology 4, 3-72, doi:10.1007/bf00630563. 1991.
- [5] A. Biryukov, M. Lamberger, F. Mendel, I. Nikolić. *Second-order differential collisions for reduced SHA-256*. Advances in Cryptology, ASIACRYPT 2011, Lecture Notes in Computer Science 7073, 270-287, doi:10.1007/978-3-642-25385-0\_15.
- [6] Yu. M. Bruck, L. G. Sodin. *On the ambiguity of the image reconstruction problem*. Optics Communications 30, 304-308, doi:10.1016/0030-4018(79)90358-4. 1979.
- [7] H. Cohn, A. Kumar, S. D. Miller, D. Radchenko, M. Viazovska. *The sphere packing problem in dimension 24*. Annals of Mathematics 185(3), doi:10.4007/annals.2017.185.3.8.
- [8] I. B. Damgård. *A design principle for hash functions*. Advances in Cryptology, CRYPTO '89, Lecture Notes in Computer Science 435, 416-427, doi:10.1007/0-387-34805-0\_39. 1990.

- [9] N. G. de Bruijn. *Algebraic theory of Penrose's non-periodic tilings of the plane, I and II*. Indagationes Mathematicae 43, 39-52 and 53-66. 1981.
- [10] P. Delsarte, J. M. Goethals, J. J. Seidel. *Spherical codes and designs*. Geometriae Dedicata 6, 363-388, doi:10.1007/bf03187604. 1977.
- [11] A. R. Edmonds. *Angular Momentum in Quantum Mechanics*. Princeton University Press. 1957.
- [12] W. Feller. *An Introduction to Probability Theory and Its Applications, Volume I, third edition*. Wiley. 1968.
- [13] National Institute of Standards and Technology. *Secure Hash Standard*. FIPS PUB 180-4, doi:10.6028/nist.fips.180-4. 2015.
- [14] P. Funk. *Beiträge zur Theorie der Kugelfunktionen*. Mathematische Annalen 77, 136-152, doi:10.1007/BF01456825. 1915.
- [15] A. Girard. *Invention nouvelle en l'algèbre*. Amsterdam; Bibliothèque nationale de France, Gallica ark:/12148/bpt6k5822034w. 1629.
- [16] G. H. Golub, C. F. Van Loan. *Matrix Computations, fourth edition*. Johns Hopkins University Press. 2013.
- [17] Á. González. *Measurement of areas on a sphere using Fibonacci and latitude-longitude lattices*. Mathematical Geosciences 42, 49-64, doi:10.1007/s11004-009-9257-x. 2010.
- [18] K. M. Górski, E. Hivon, A. J. Banday, B. D. Wandelt, F. K. Hansen, M. Reinecke, M. Bartelmann. *HEALPix: a framework for high-resolution discretization and fast analysis of data distributed on the sphere*. The Astrophysical Journal 622, 759-771, doi:10.1086/427976. 2005.
- [19] L. Greengard, V. Rokhlin. *A fast algorithm for particle simulations*. Journal of Computational Physics 73, 325-348, doi:10.1016/0021-9991(87)90140-9. 1987.
- [20] N. A. Gumerov, R. Duraiswami. *Recursive computation of spherical harmonic rotation coefficients of large degree*. arXiv:1403.7698. 2014.
- [21] T. C. Hales. *A proof of the Kepler conjecture*. Annals of Mathematics 162(3) 1065-1185, doi:10.4007/annals.2005.162.1065.
- [22] M. H. Hayes. *The reconstruction of a multidimensional sequence from the phase or magnitude of its Fourier transform*. IEEE Transactions on Acoustics, Speech, and Signal Processing 30, 140-154, doi:10.1109/tassp.1982.1163863.
- [23] T. L. Heath. *The Works of Archimedes*. Cambridge University Press. 1897.
- [24] E. Hecke. *Über orthogonal-invariante Integralgleichungen*. Mathematische Annalen 78, 398-404, doi:10.1007/BF01457114. 1917.

- [25] S. A. Holmes, W. E. Featherstone. *A unified approach to the Clenshaw summation and the recursive computation of very high degree and order normalised associated Legendre functions*. Journal of Geodesy 76, 279-299, doi:10.1007/s00190-002-0216-2. 2002.
- [26] M. Kazhdan, T. Funkhouser, S. Rusinkiewicz. *Rotation invariant spherical harmonic representation of 3D shape descriptors*. Symposium on Geometry Processing, 156-165, doi:10.2312/SGP/SGP03/156-165. 2003.
- [27] D. Khovratovich, C. Rechberger, A. Savelieva. *Bicliques for preimages: attacks on Skein-512 and the SHA-2 family*. Fast Software Encryption 2012, Lecture Notes in Computer Science 7549, 244-263, doi:10.1007/978-3-642-34047-5\_15.
- [28] J. M. Lee, G. Uhlmann. *Determining anisotropic real-analytic conductivities by boundary measurements*. Communications on Pure and Applied Mathematics 42, 1097-1112, doi:10.1002/cpa.3160420804. 1989.
- [29] D. A. Levin, Y. Peres, with contributions by E. L. Wilmer. *Markov Chains and Mixing Times, second edition*. American Mathematical Society. 2017.
- [30] D. A. Levin, Y. Peres, E. L. Wilmer. *Markov Chains and Mixing Times*. American Mathematical Society. 2009.
- [31] Yingxin Li, Fukang Liu, Gaoli Wang. *New Records in Collision Attacks on SHA-2*. IACR ePrint 2024/349.  
Copy held sources/computer\_science/cryptography/eprint\_2024\_349.pdf, SHA-256  
ca79053516e9110d0e1c99a1f0638f1862d43372ba64e5ba9c50b85631e8ce05.
- [32] Y. Li, F. Liu, G. Wang, X. Dong, S. Sun. *The first practical collision for 31-step SHA-256*. Advances in Cryptology, ASIACRYPT 2024, doi:10.1007/978-981-96-0941-3\_8.
- [33] N. Mandache. *Exponential instability in an inverse problem for the Schrödinger equation*. Inverse Problems 17, 1435-1444, doi:10.1088/0266-5611/17/5/313. 2001.
- [34] M. Matsui. *Linear cryptanalysis method for DES cipher*. Advances in Cryptology, EUROCRYPT '93, Lecture Notes in Computer Science 765, 386-397, doi:10.1007/3-540-48285-7\_33. 1994.
- [35] Florian Mendel, Tomislav Nad, Martin Schläffer. *Improving Local Collisions: New Attacks on Reduced SHA-256*. IACR ePrint 2015/350.  
Copy held sources/computer\_science/cryptography/eprint\_2015\_350.pdf, SHA-256  
e116097f8848641ff76b9c41630069bb5b529ba6d24df5f386b25c1c5e42e6e4.
- [36] R. C. Merkle. *One way hash functions and DES*. Advances in Cryptology, CRYPTO '89, Lecture Notes in Computer Science 435, 428-446, doi:10.1007/0-387-34805-0\_40. 1990.
- [37] Peter J. Mohr, David B. Newell, Barry N. Taylor, Eite Tiesinga. *CODATA recommended values of the fundamental physical constants: 2022*. doi:10.1103/RevModPhys.97.025002. 2025.  
Copy held sources/physics/metrology/revmodphys\_97\_025002.pdf, SHA-256  
e7fa786c5b66640784451f40883c886bddaee52d87521382af71c55e1636af41.

- [38] R. Ramamoorthi, P. Hanrahan. *An efficient representation for irradiance environment maps*. Proceedings of SIGGRAPH 2001, 497-500, doi:10.1145/383259.383317.
- [39] A. Rényi. *On measures of entropy and information*. Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability 1, 547-561. 1961.
- [40] V. A. Rokhlin. *Lectures on the entropy theory of measure-preserving transformations*. Russian Math. Surveys 22(5):1-52, Math-Net.Ru rm5788. 1967.  
Copy held sources/mathematics/ergodic\_theory/rm5788\_rokhlin\_1967.pdf, SHA-256  
b86f10f07848830a85fbacf65dbe3f03d0b62138776d7c227efafee251818dbd.
- [41] W. F. Sheppard. *On the application of the theory of error to cases of normal distribution and normal correlation*. Philosophical Transactions of the Royal Society of London A 192, 101-167, doi:10.1098/rsta.1899.0003.
- [42] L. Susskind. *The world as a hologram*. Journal of Mathematical Physics 36, 6377-6396, doi:10.1063/1.531249. 1995.
- [43] S. Świerczkowski. *On successive settings of an arc on the circumference of a circle*. Fundamenta Mathematicae 46, 187-189, doi:10.4064/fm-46-2-187-189. 1958.
- [44] R. Swinbank, R. J. Purser. *Fibonacci grids: a novel approach to global modelling*. Quarterly Journal of the Royal Meteorological Society 132, 1769-1793, doi:10.1256/qj.05.227. 2006.
- [45] M. S. Viazovska. *The sphere packing problem in dimension 8*. Annals of Mathematics 185(3), doi:10.4007/annals.2017.185.3.7.
- [46] H. Vogel. *A better way to construct the sunflower head*. Mathematical Biosciences 44, 179-189, doi:10.1016/0025-5564(79)90080-4. 1979.
- [47] A. Wald, J. Wolfowitz. *On a test whether two samples are from the same population*. Annals of Mathematical Statistics 11, 147-162, doi:10.1214/aoms/1177731909. 1940.
- [48] A. Walfisz. *Gitterpunkte in mehrdimensionalen Kugeln*. Monografie Matematyczne 33, Państwowe Wydawnictwo Naukowe, Warsaw. 1957.
- [49] A. F. Webster, S. E. Tavares. *On the design of S-boxes*. Advances in Cryptology, CRYPTO '85, Lecture Notes in Computer Science 218, 523-534, doi:10.1007/3-540-39799-x\_41. 1986.
- [50] H. Weyl. *Über die asymptotische Verteilung der Eigenwerte*. Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse, 110-117. 1911.
- [51] C. A. White, M. Head-Gordon. *Rotating around the quartic angular momentum barrier in fast multipole method calculations*. Journal of Chemical Physics 105, 5061-5067, doi:10.1063/1.472369. 1996.
- [52] L. Williams. *Pyramidal parametrics*. Computer Graphics 17(3), 1-11, doi:10.1145/964967.801126. 1983.
- [53] D. H. Wolpert. *Physical limits of inference*. Physica D 237, 1257-1281, doi:10.1016/j.physd.2008.03.040.

- [54] R. S. Womersley, I. H. Sloan. *How good can polynomial interpolation on the sphere be?*. Advances in Computational Mathematics 14, 195-226, doi:10.1023/a:1016630227163. 2001.